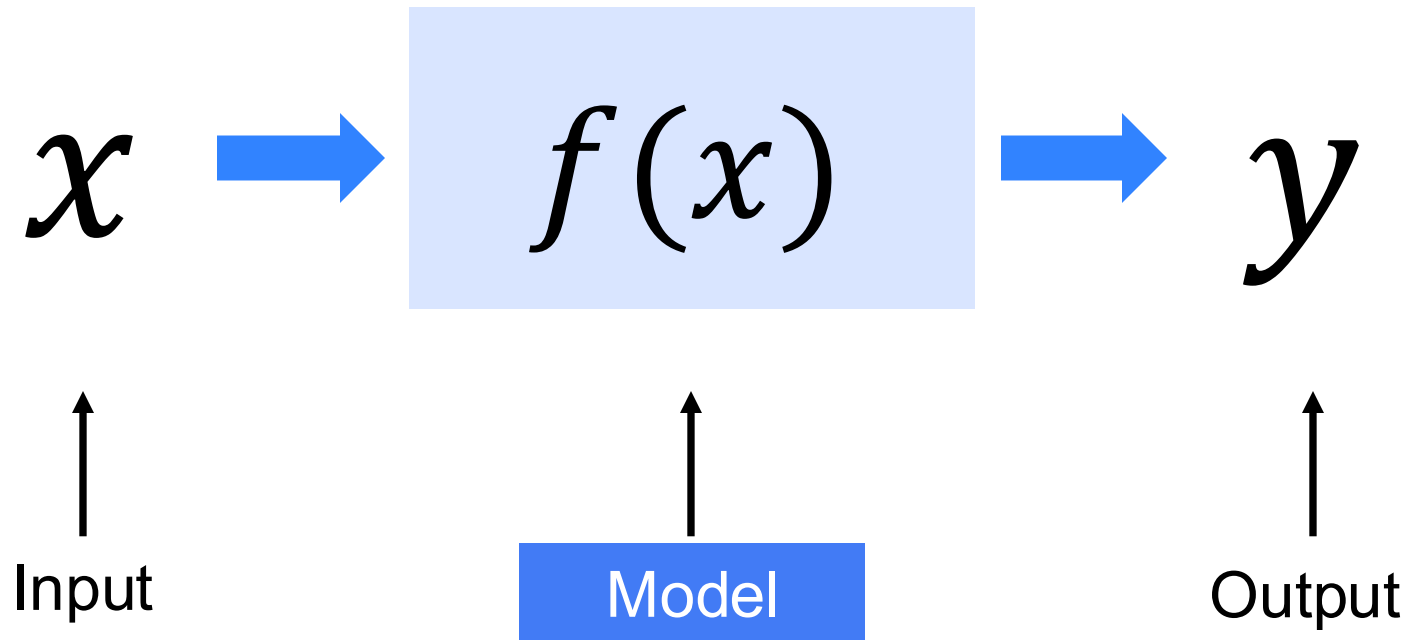


```
94 ...len($hex_str) == 6 ) {
95     $color_val = hexdec($hex_str);
96     $rgb_array['r'] = 0xFF & ($color_val >> 0x10);
97     $rgb_array['g'] = 0xFF & ($color_val >> 0x8);
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105 }
106 }
107 }
108 // Draw
109 ...return_string ? implode...
```

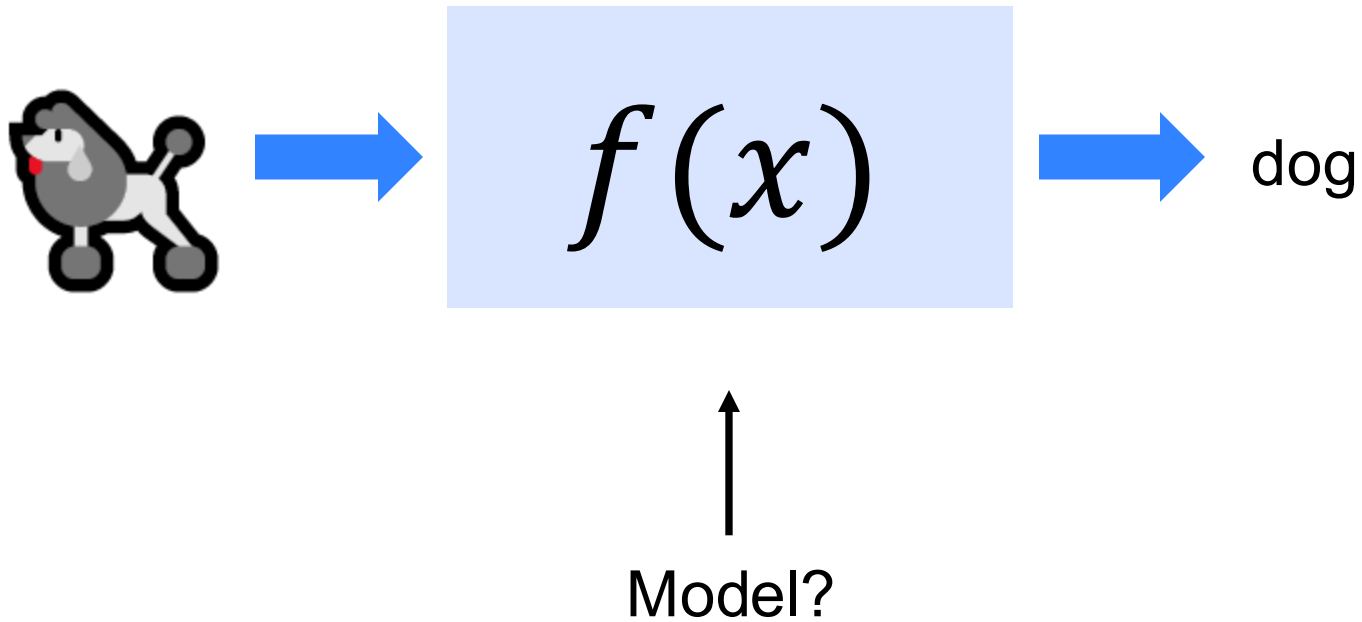
Artificial Intelligence in Interactive Systems

Supervised vs. Unsupervised Learning

Introduction into Machine Learning



Introduction into Machine Learning



*object recognition

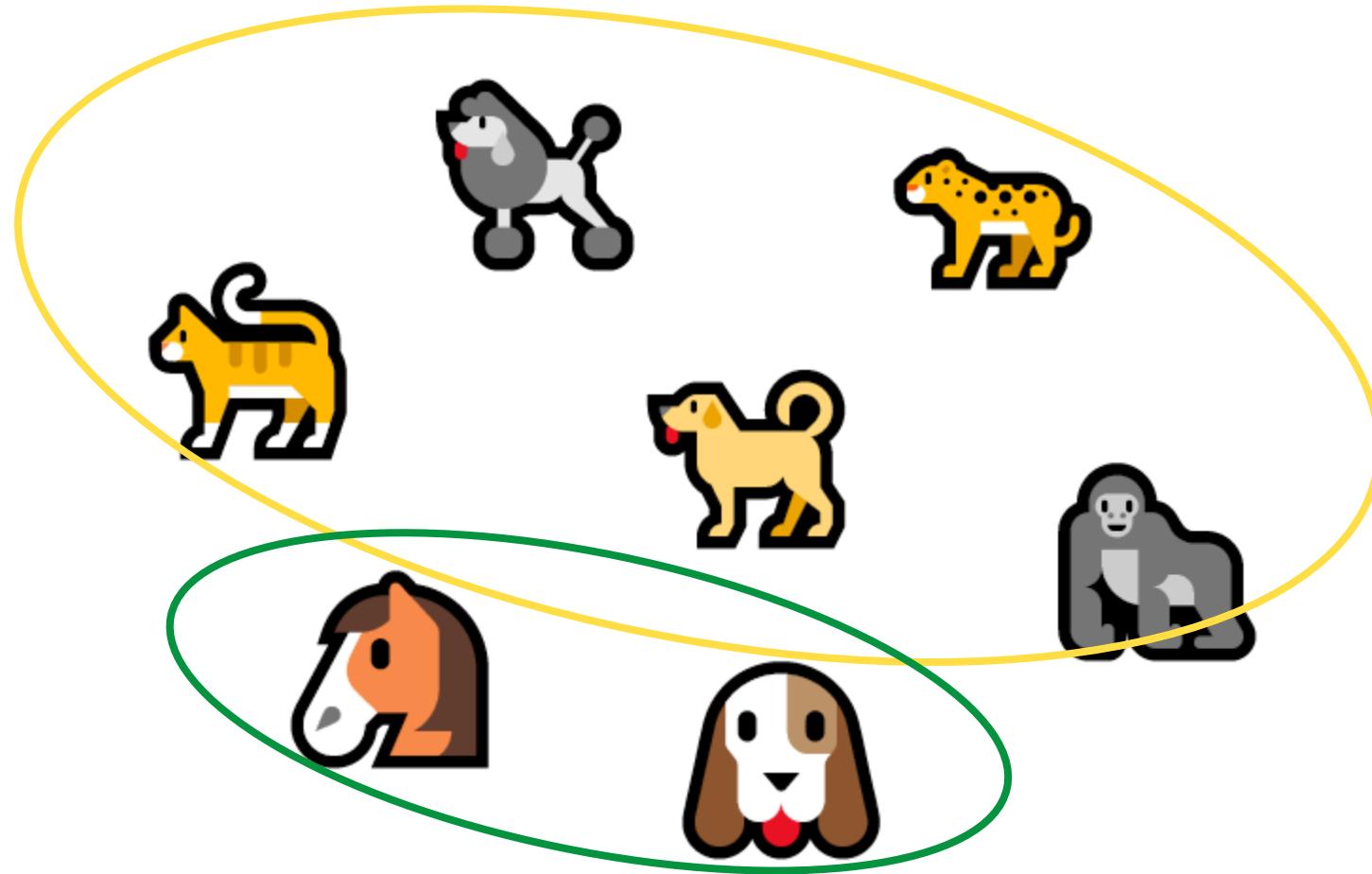
Types of ML Problems

What can we learn just by looking at the data?



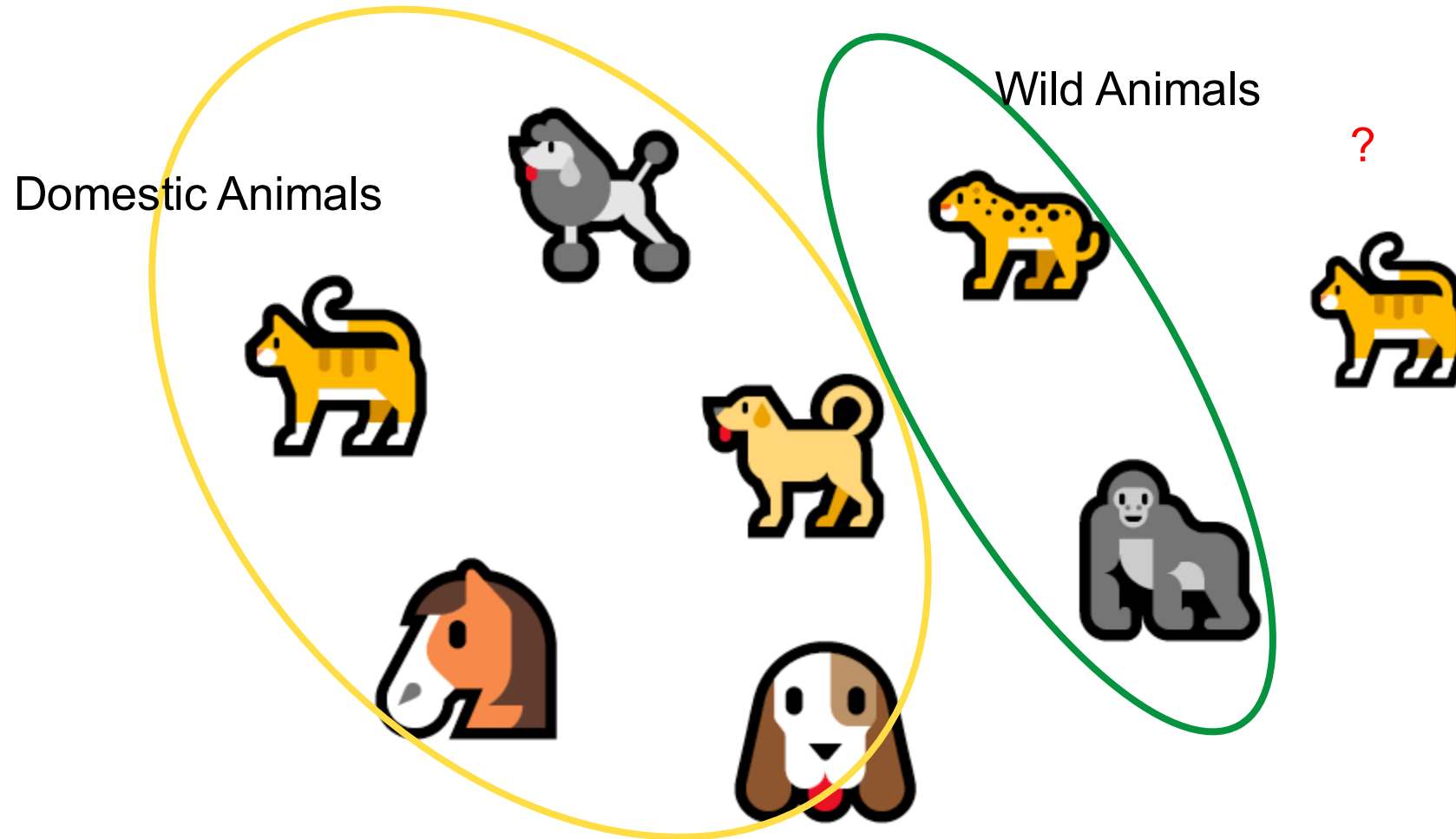
Types of ML Problems

What can we learn just by looking at the data?



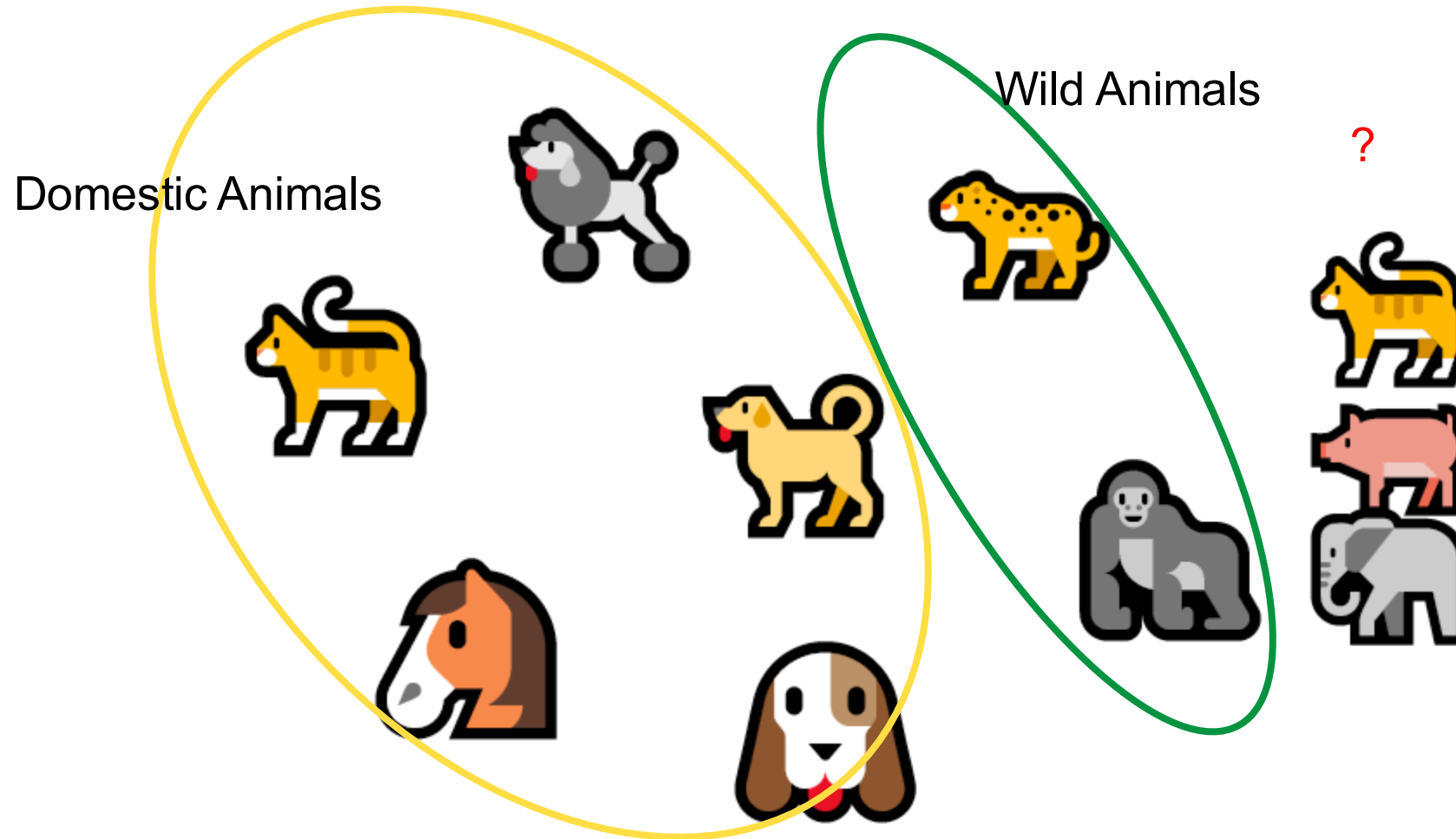
Types of ML Problems

What can we do with binned data?



Types of ML Problems

What can we do with binned data?



Learning Strategies

Supervised
Learning

Unsupervised
Learning

Learning Strategies

Supervised
Learning

Unsupervised
Learning

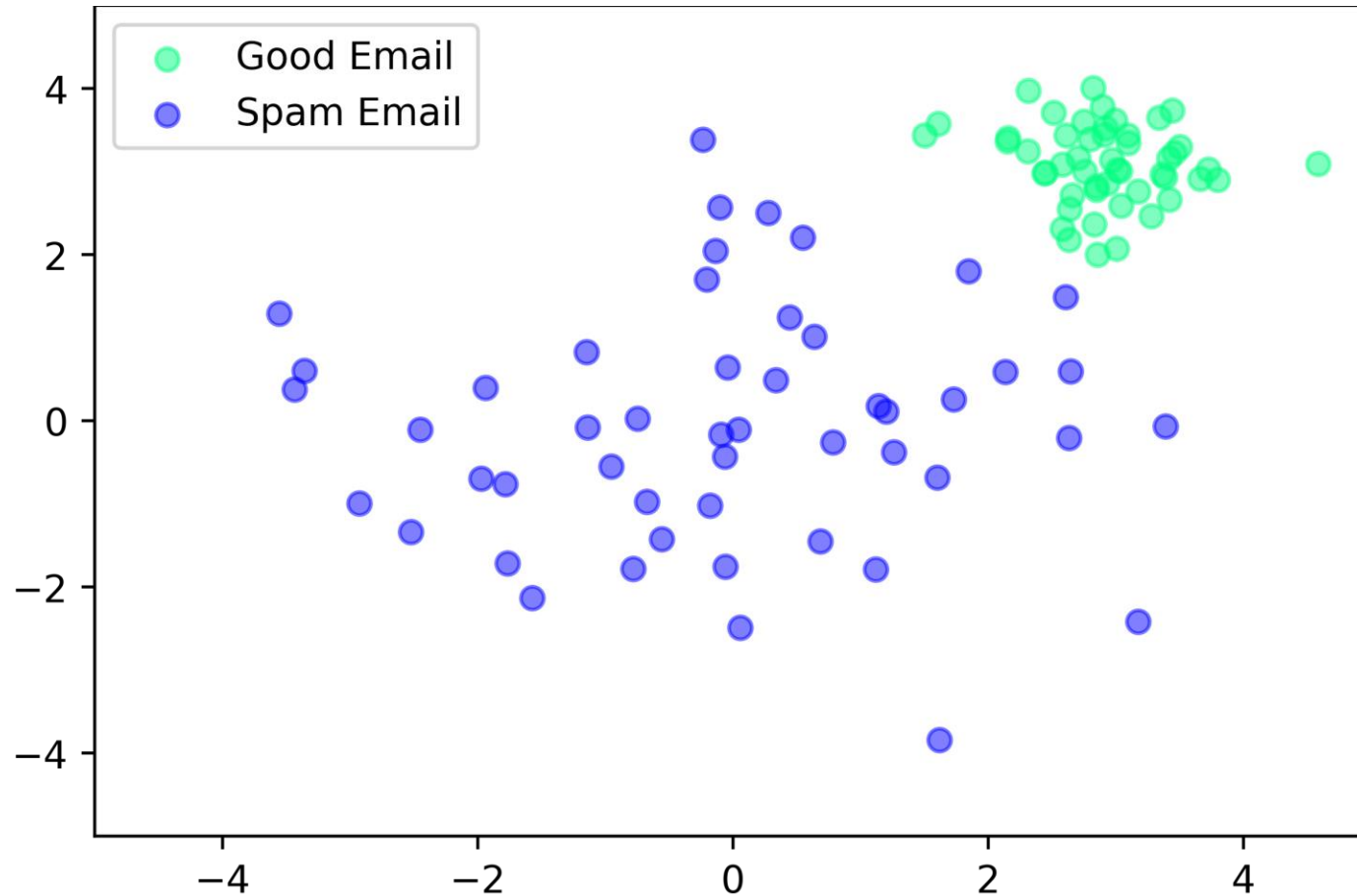
Reinforcement
learning

Learning Strategies

	Supervised Learning	Unsupervised Learning
Discrete	Classification or Categorization	Clustering
Continuous	Regression	Dimensionality reduction

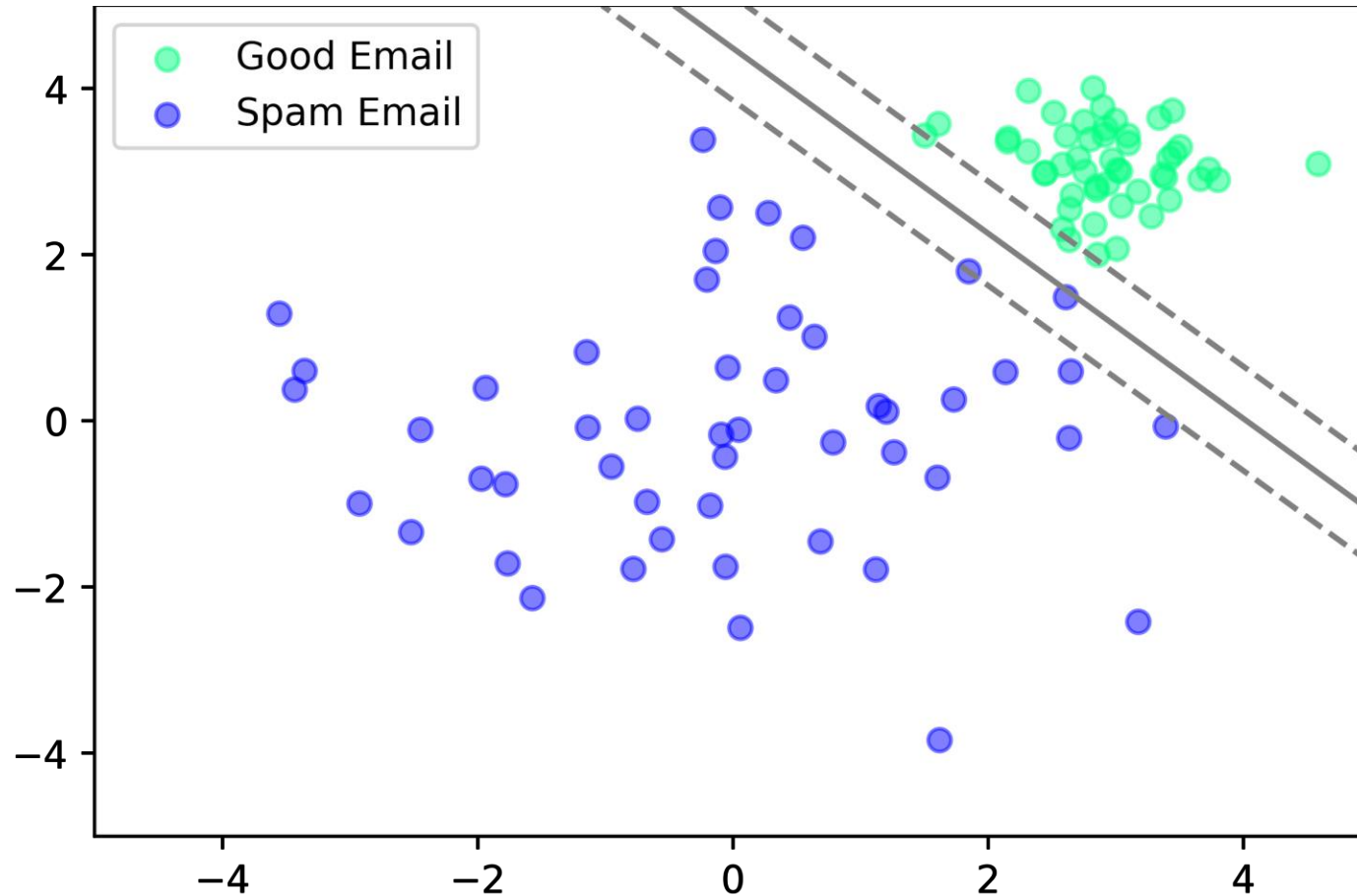
Supervised Learning

SVM Example



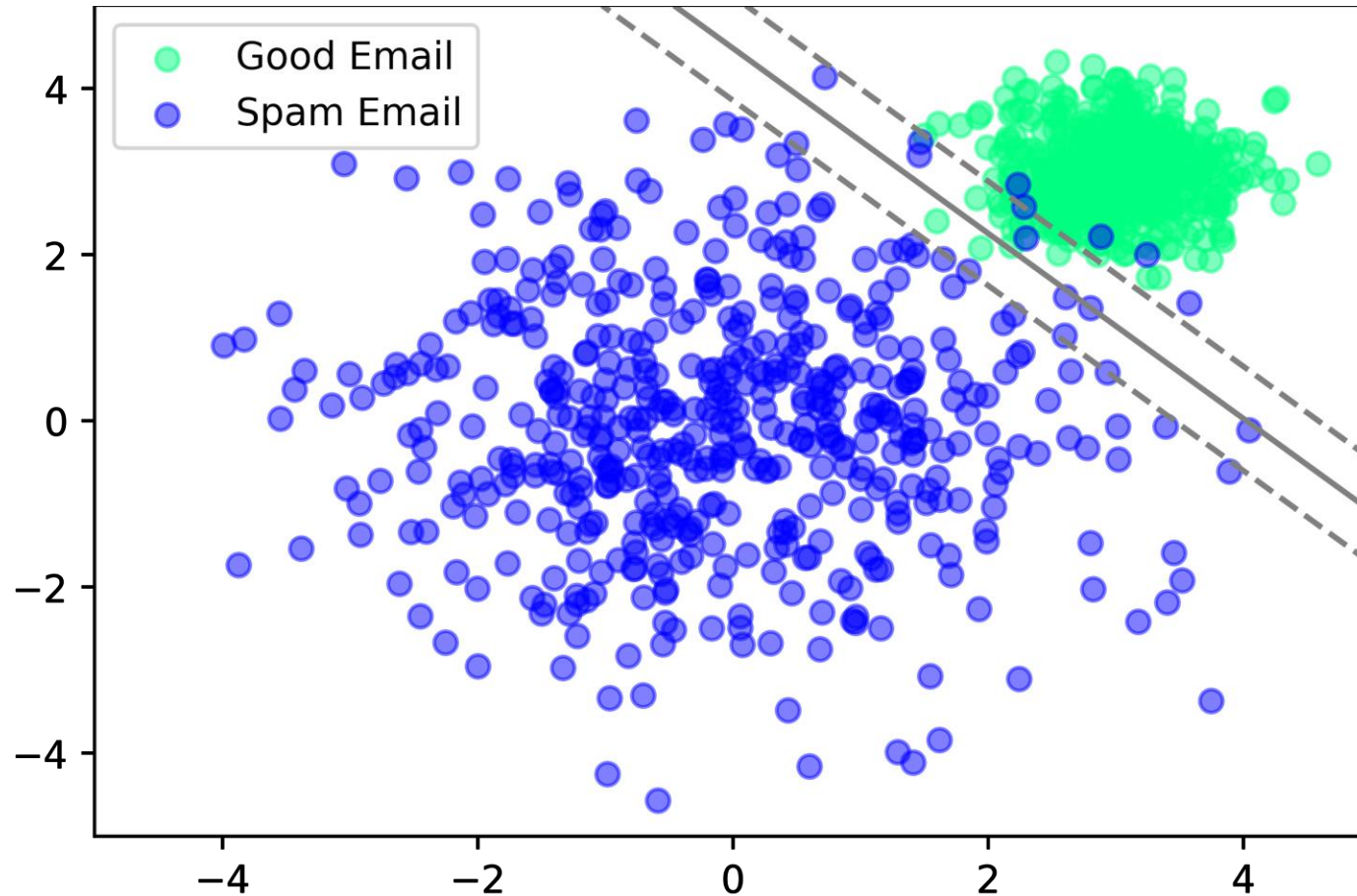
Supervised Learning

SVM Example

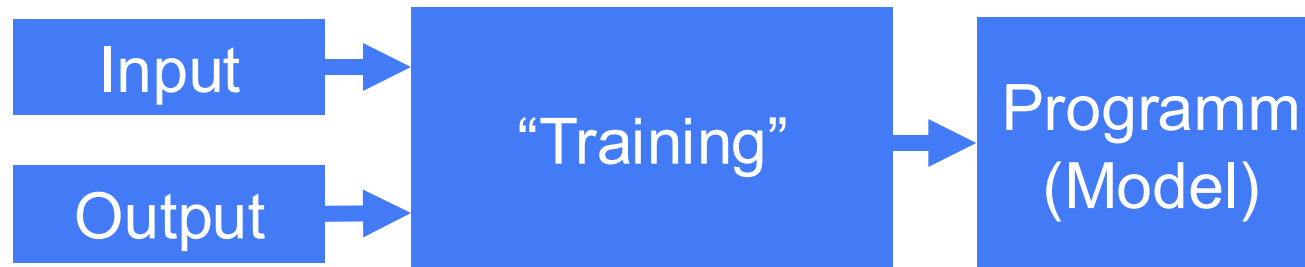


Supervised Learning

SVM Example



Supervised Learning



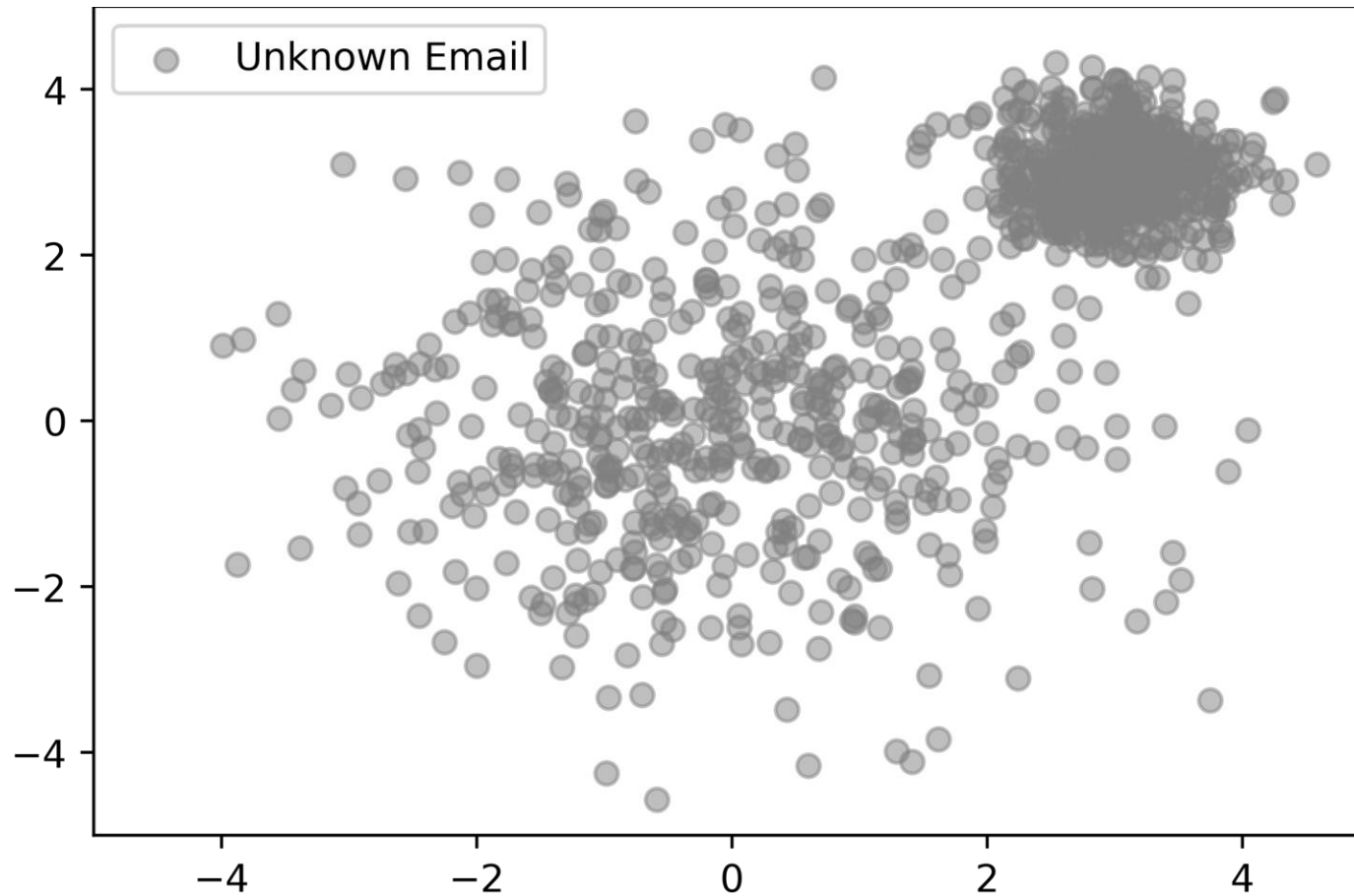
Supervised Learning

- Data

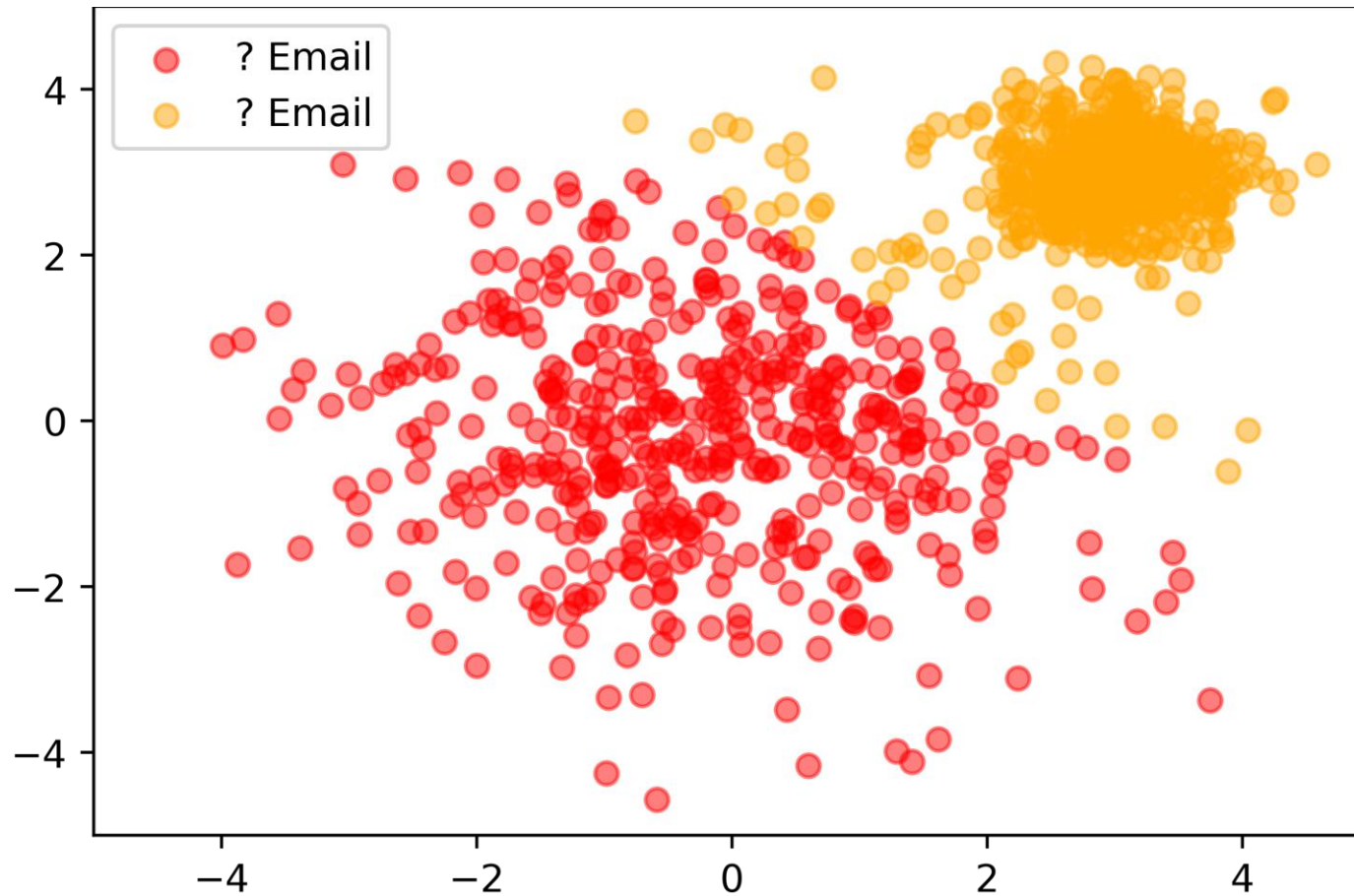


- Labels (Classes)
- Continues values

Unsupervised Learning



Unsupervised Learning



Unsupervised Learning

- Data



Not needed:

- Labels (Classes)
- Continuous values

Learning Strategies

	Supervised Learning	Unsupervised Learning
Discrete	Classification or Categorization	Clustering
Continuous	Regression	Dimensionality reduction

Conclusion

Supervised vs. Unsupervised Learning

- Unsupervised Learning
 - Unlabeled data
 - Use to discover patterns and information
- Supervised Learning
 - Labeled data
 - Used to generalize known relations between input and output

```
94 // ...
95 $color_val = hexdec($hex_str);
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108 // Draw
109
```

Practical Machine Learning

Supervised Learning

Learning Strategies

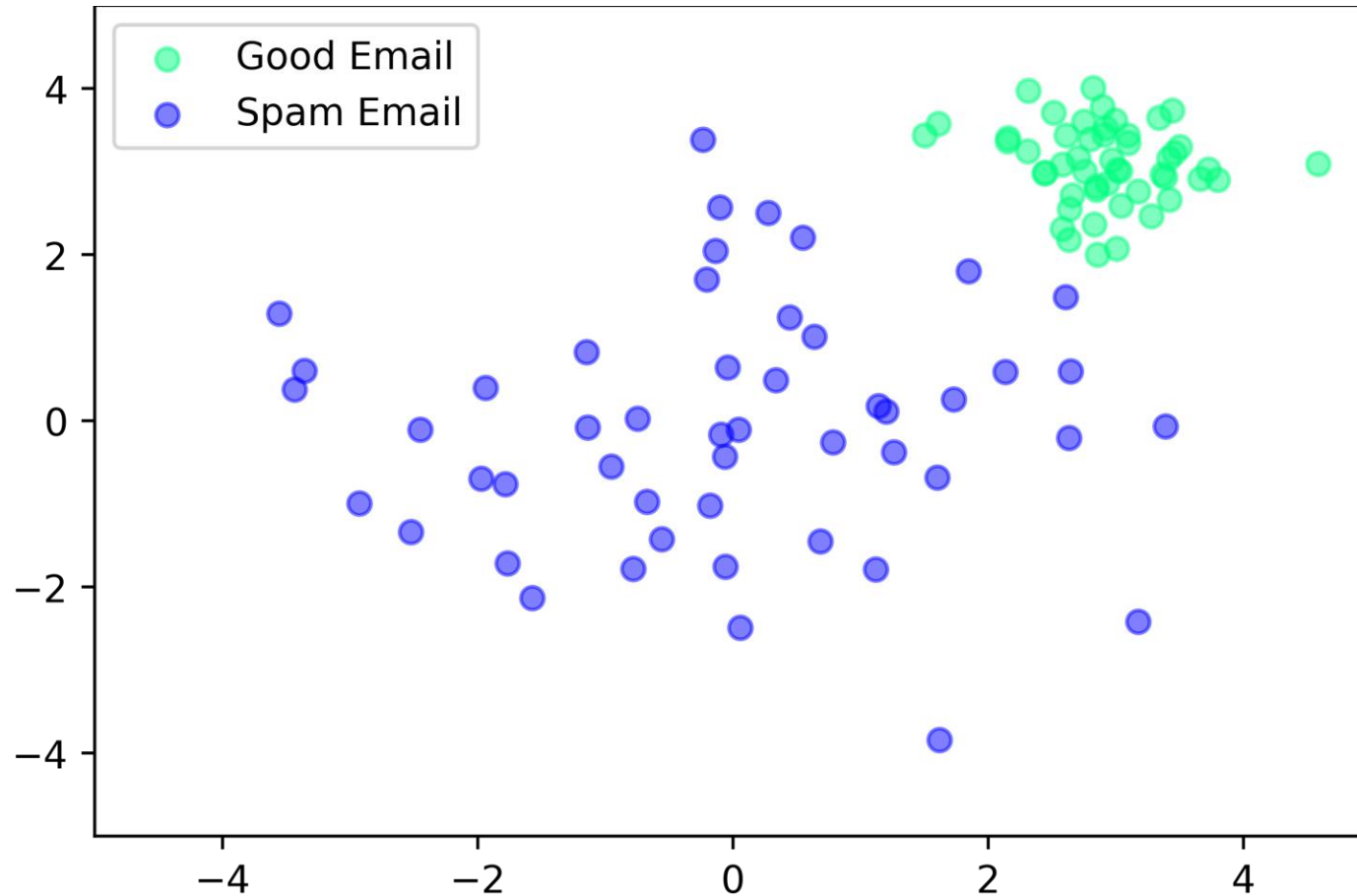
	Supervised Learning	Unsupervised Learning
Discrete	Classification or Categorization	Clustering
Continuous	Regression	Dimensionality reduction

Classification

Supervised Learning

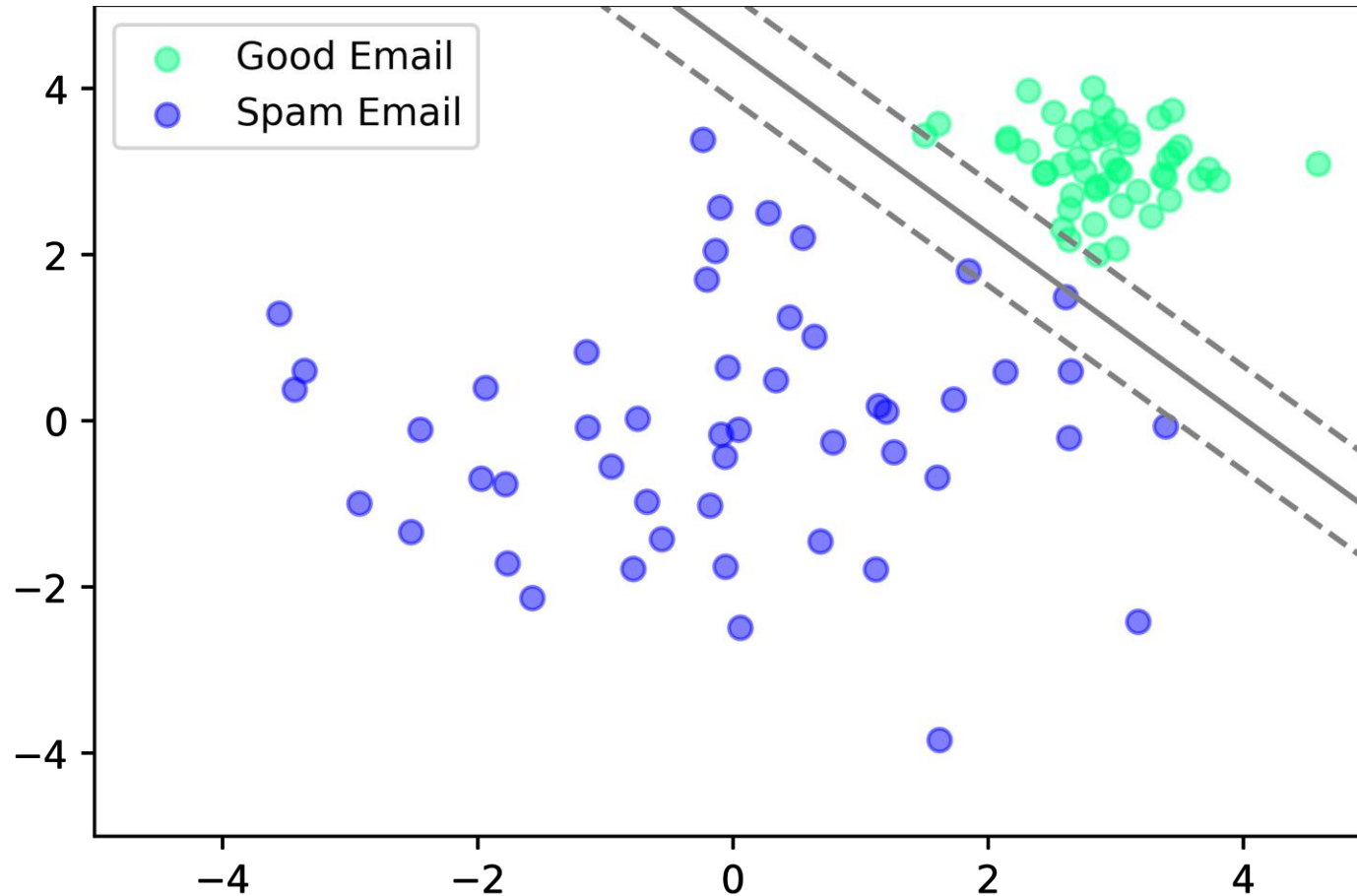
Classification

SVM Example



Classification

SVM Example



How? One-Hot-Encoded

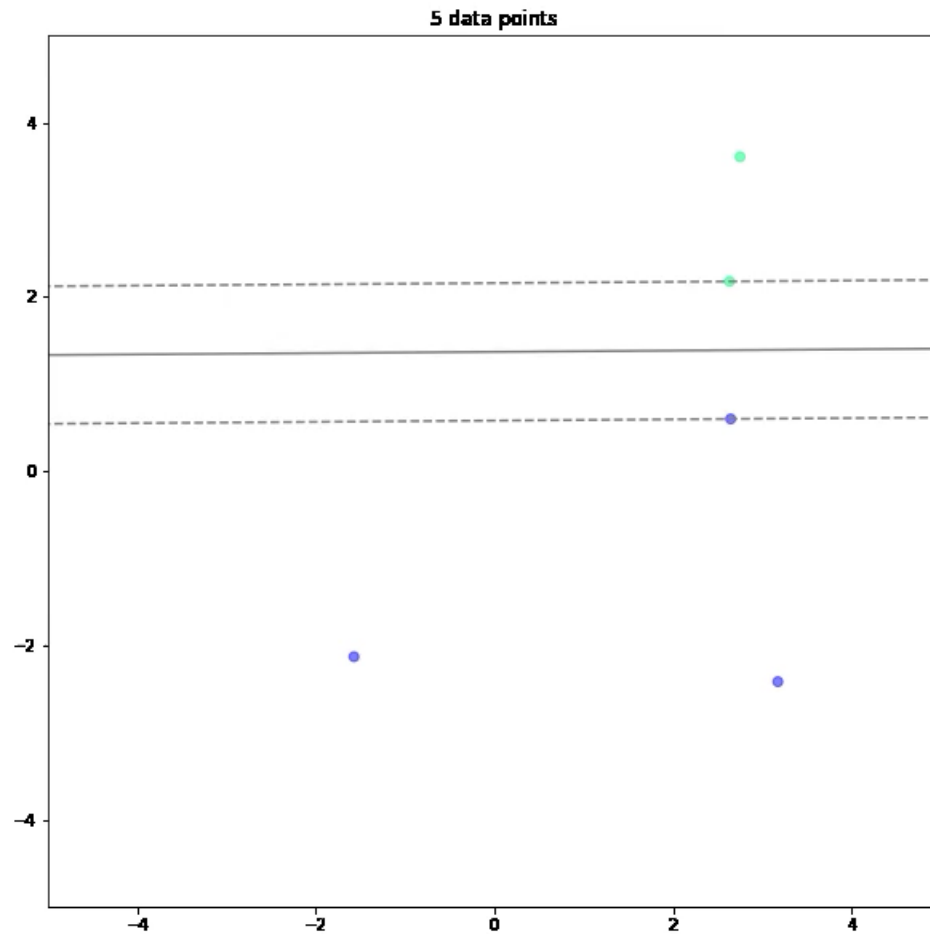
Supervised Learning

Classification



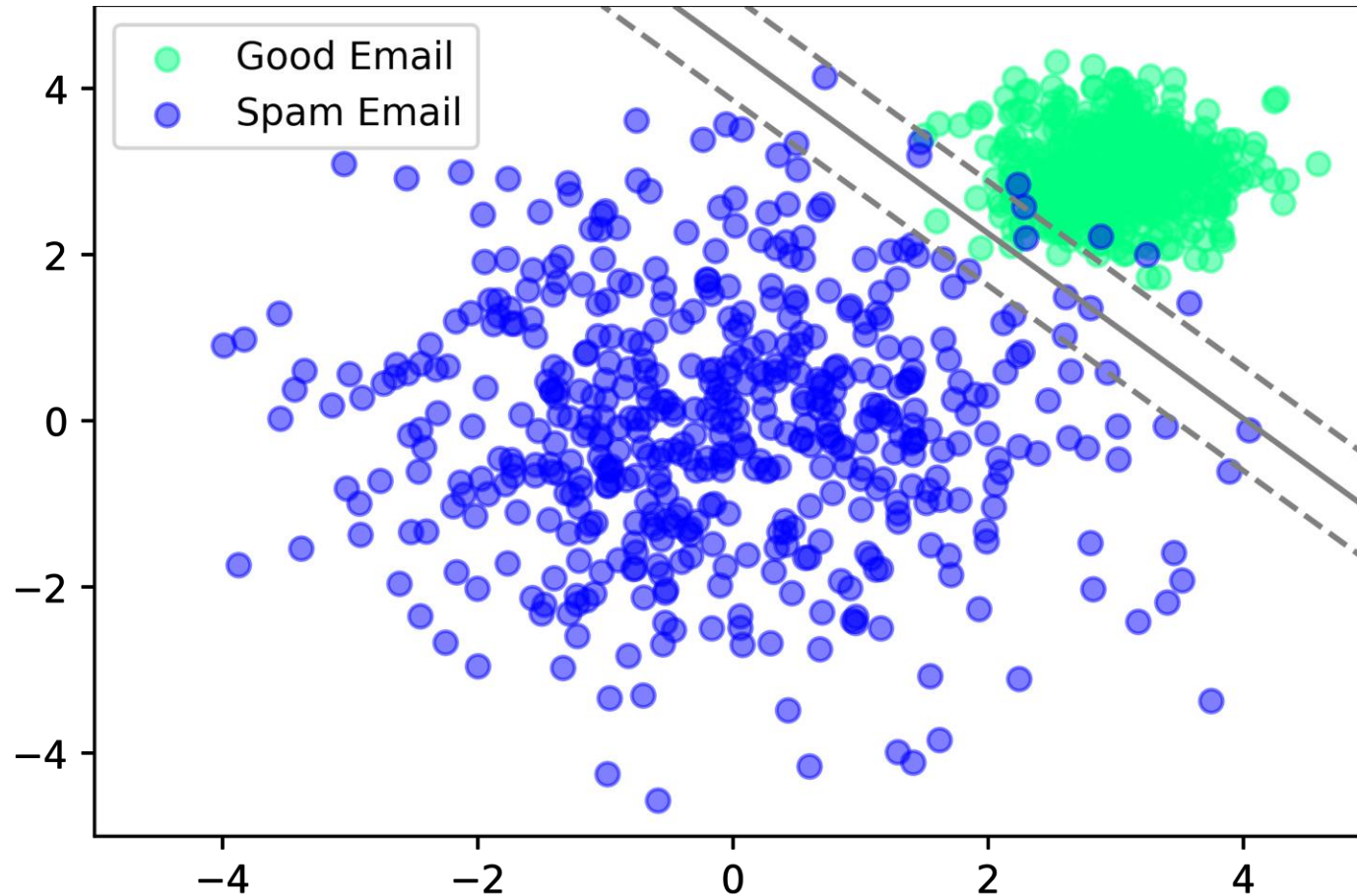
Classification

SVM Example



Classification

SVM Example



Regression

Supervised Learning

Regression

Classification

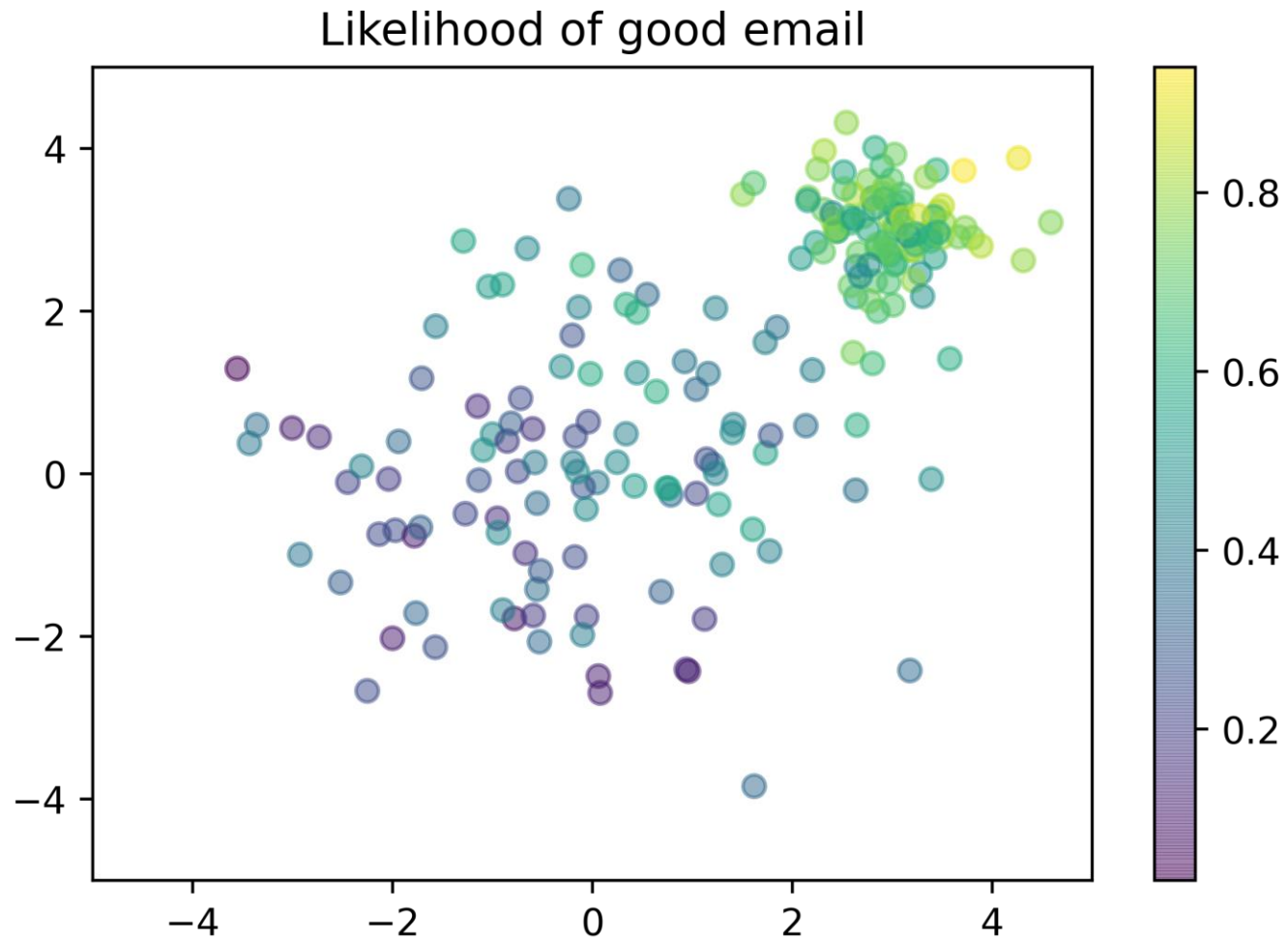
$$\begin{bmatrix} 2.1 \\ 1.8 \end{bmatrix} \Rightarrow \textit{good}$$

Regression

$$\begin{bmatrix} 2.1 \\ 1.8 \end{bmatrix} \Rightarrow .9$$

With a likelihood of
90% is this email good.

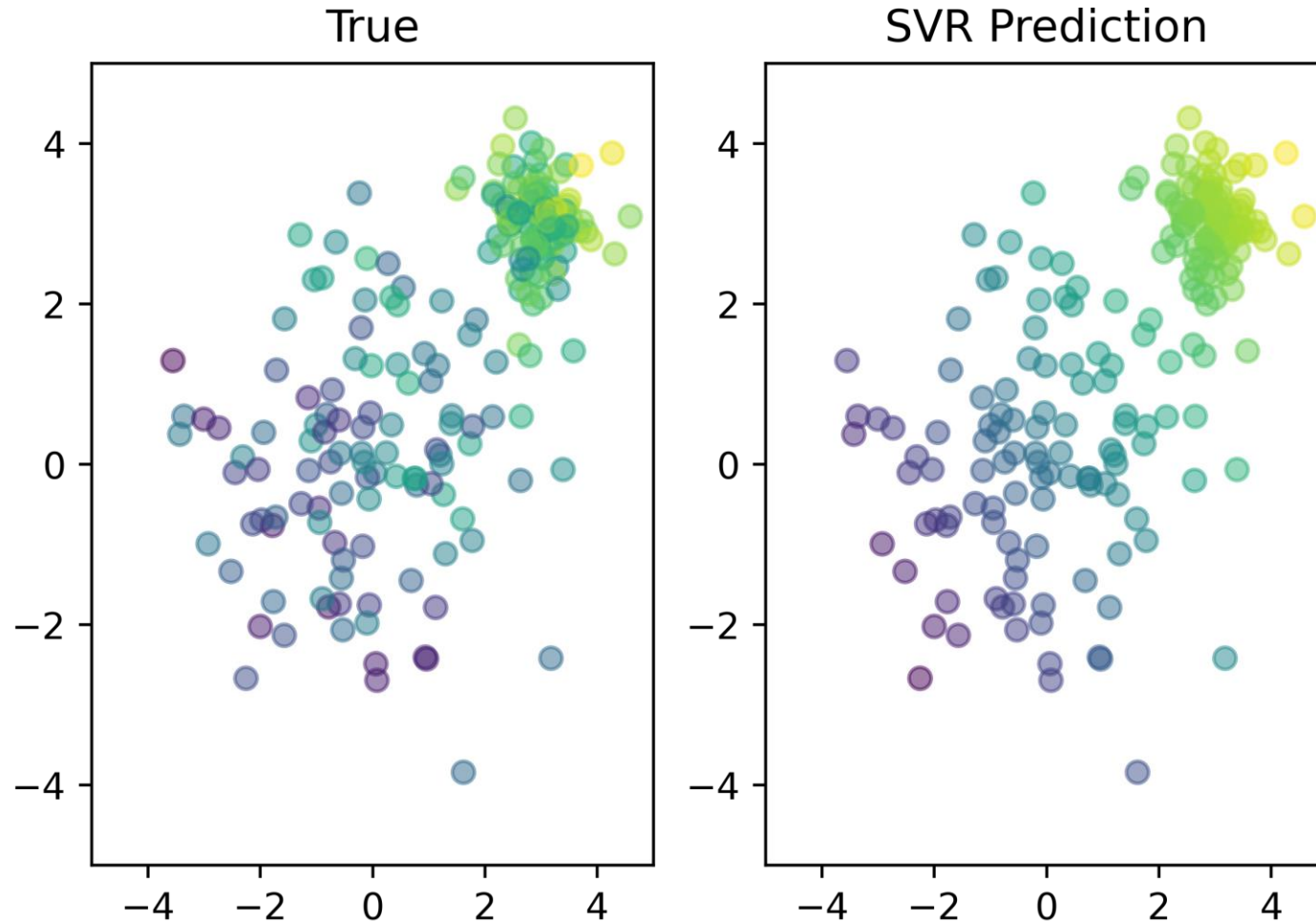
Regression



How? Float value.

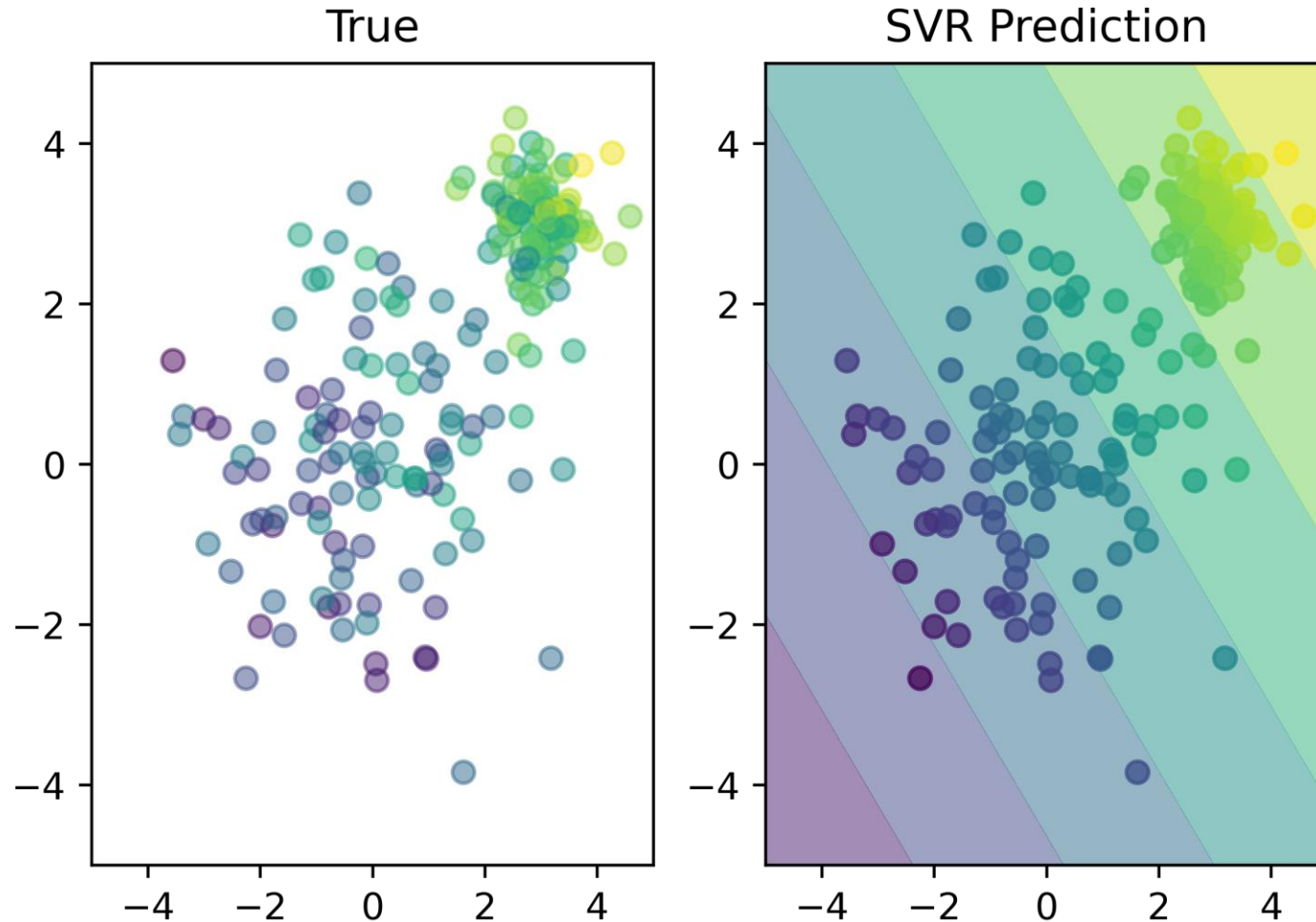
Regression

Linear Support Vector Regression (SVR)



Regression

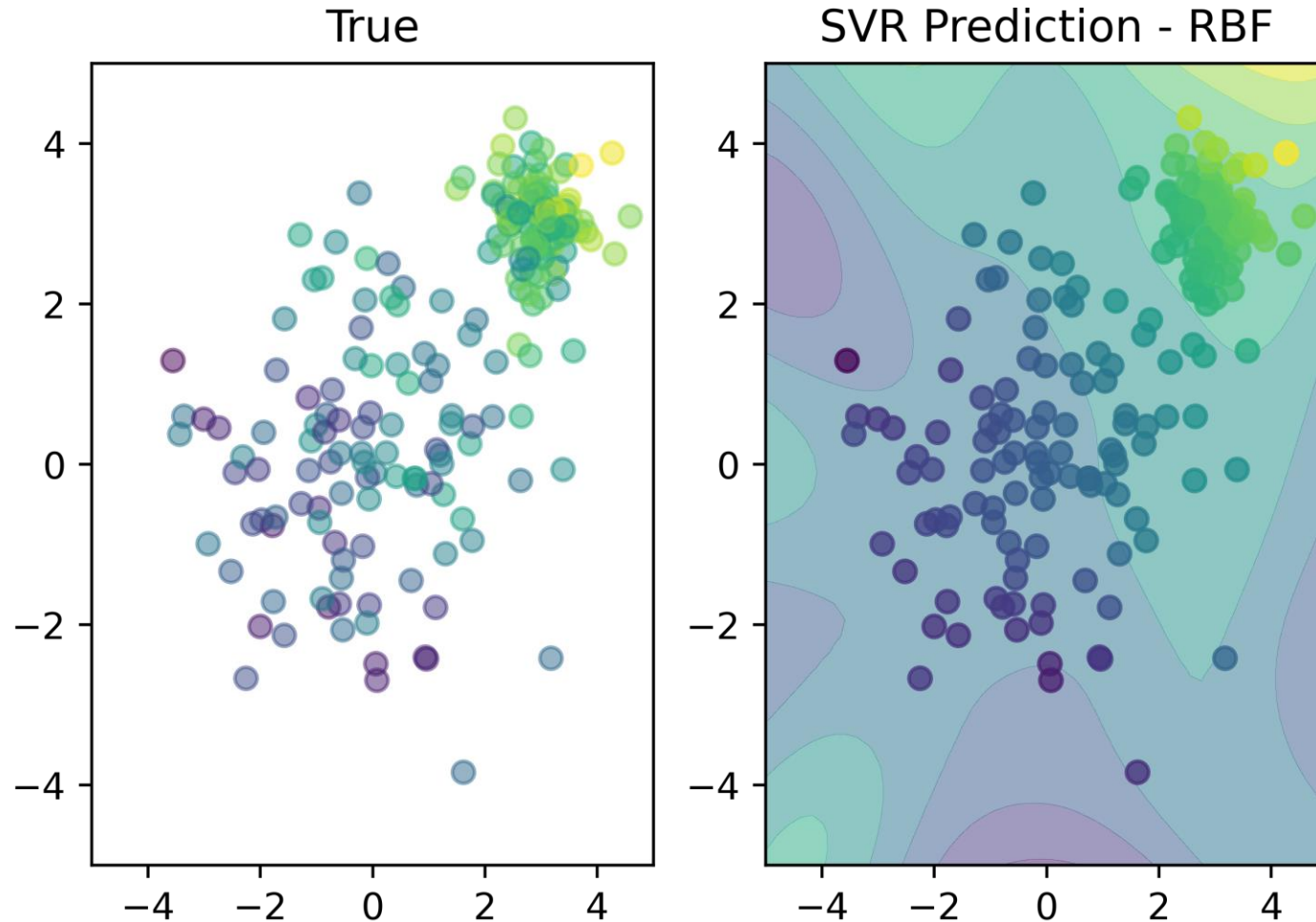
Linear Support Vector Regression (SVR)



What is an SVR in a 2D space?

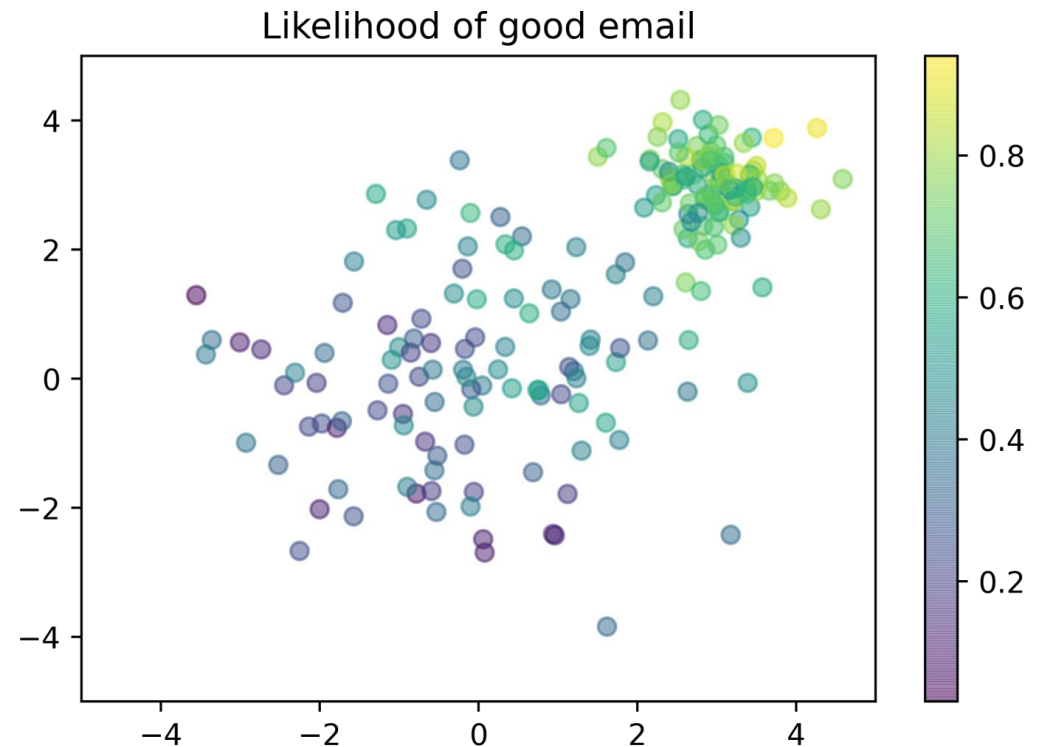
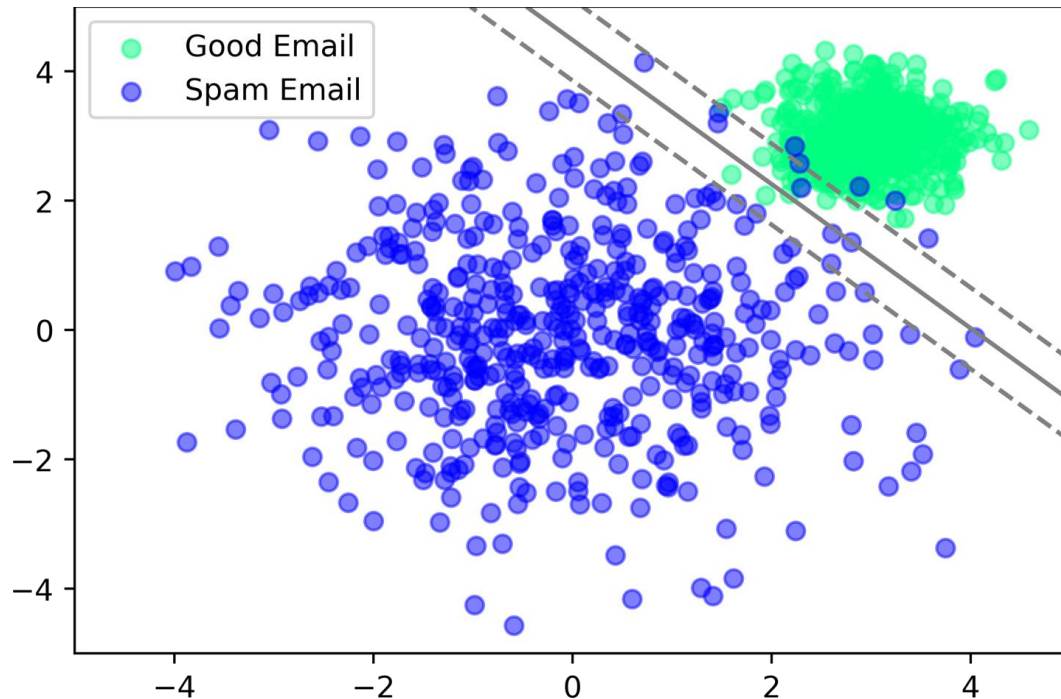
Regression

RBF Support Vector Regression (SVR)



What is an SVR in a 2D space?

Transforming Regression to Classification?



How? Quantization – Quality?

Conclusion

Supervised Learning

- Classification
 - Labeled data (paired input and label)
 - The label is of a class (e.g., dog, [grey, dog])
- Regression
 - Labeled data (paired input and label)
 - Labels are on a continuous scale (e.g., 3.4, [4.5, -17.1])

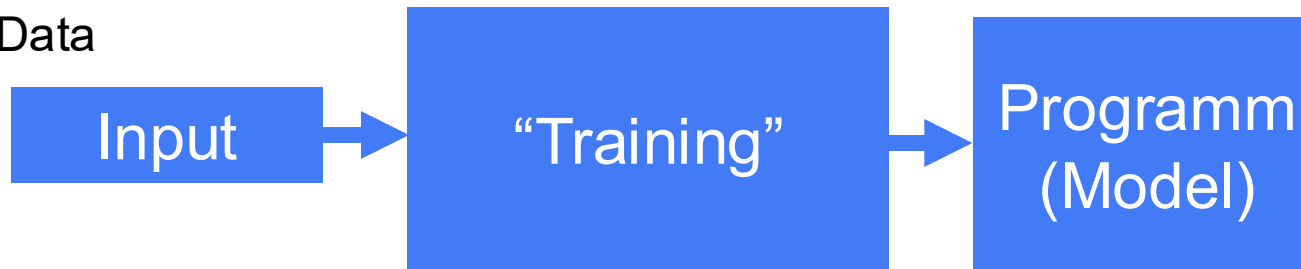
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103 } else {
104     return false;
105 }
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107 }
108 // Draw
109 ...return_string ? implode(
```

Practical Machine Learning

Unsupervised Learning

Unsupervised Learning

- Data

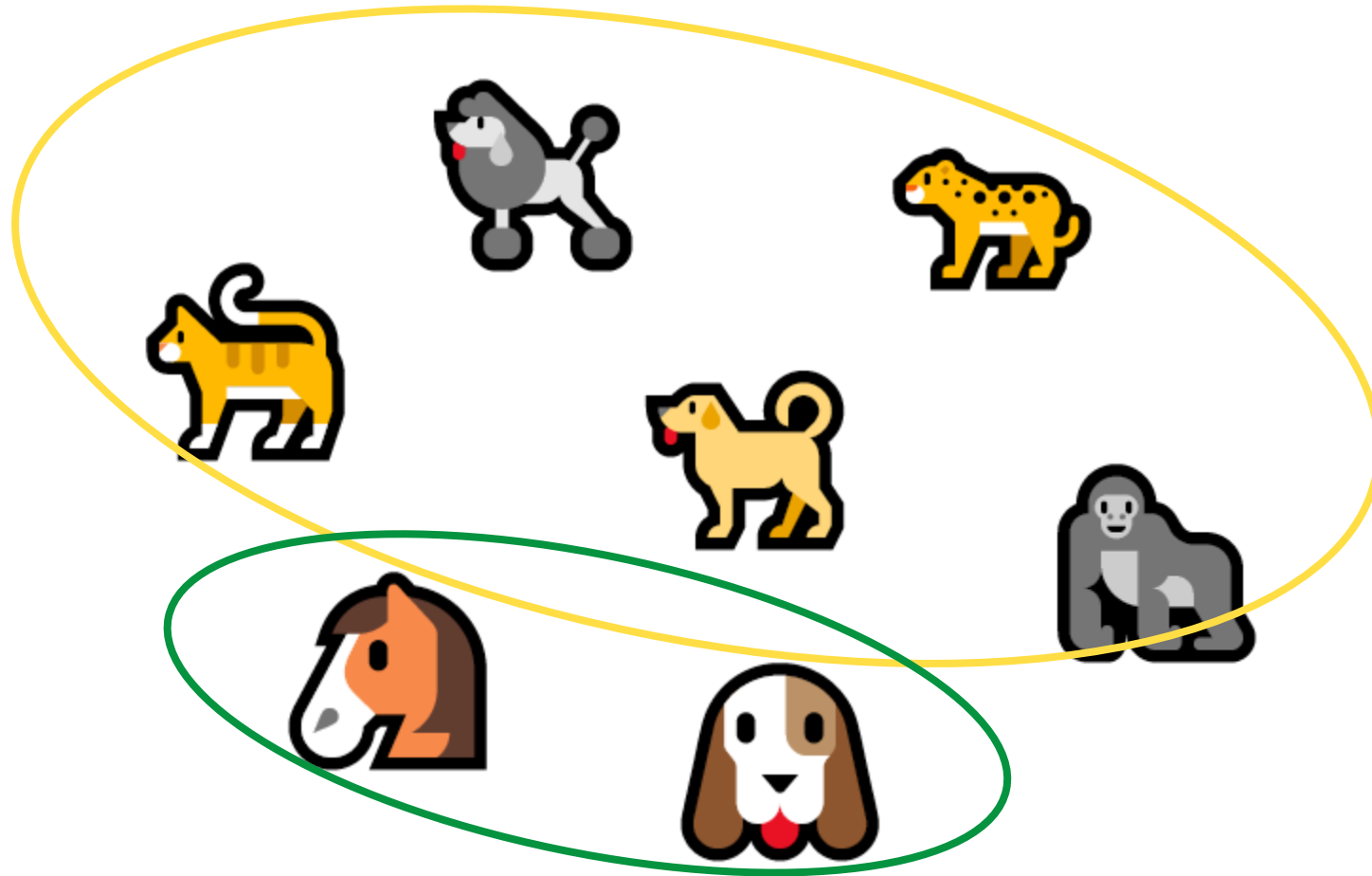


Not needed:

- Labels (Classes)
- Continuous values

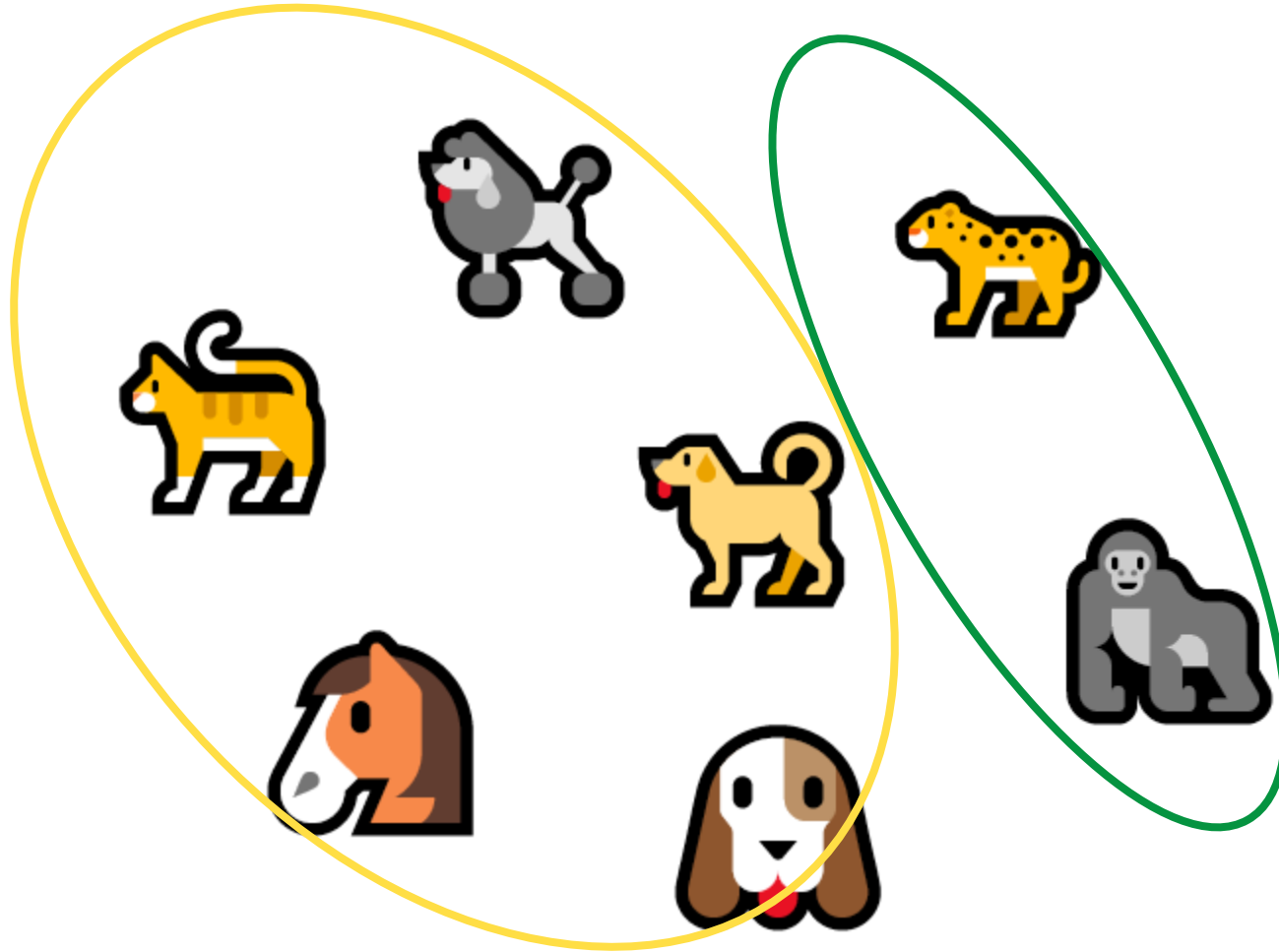
Unsupervised Learning

What can we learn just by looking at the data?



Unsupervised Learning

What can we learn just by looking at the data?



Learning Strategies

	Supervised Learning	Unsupervised Learning
Discrete	Classification or Categorization	Clustering
Continuous	Regression	Dimensionality reduction

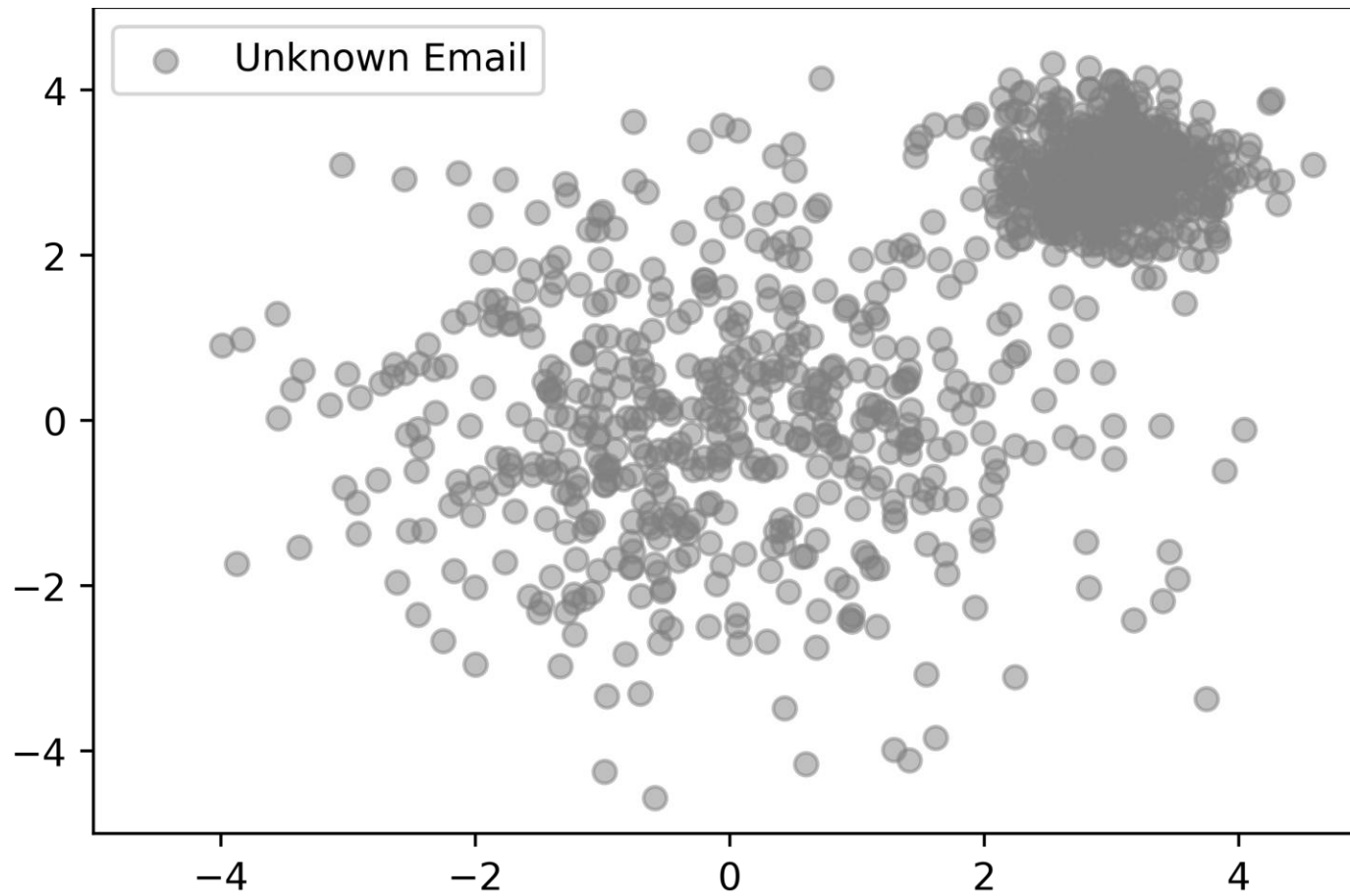
Unsupervised Learning Methods

- Hierarchical clustering
- **K-means clustering**
- **Principal Component Analysis (PCA)**
- Singular Value Decomposition
- Independent Component Analysis
-

Clustering

Unsupervised Learning

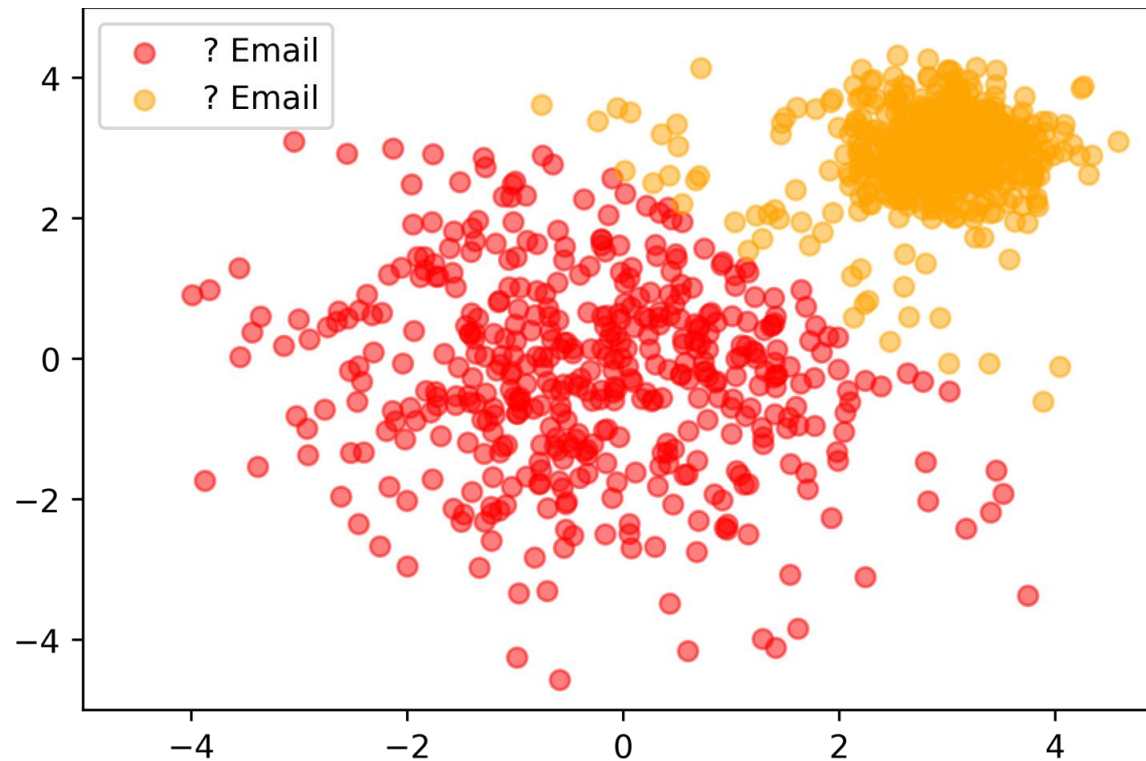
Unknown Data



Clustering

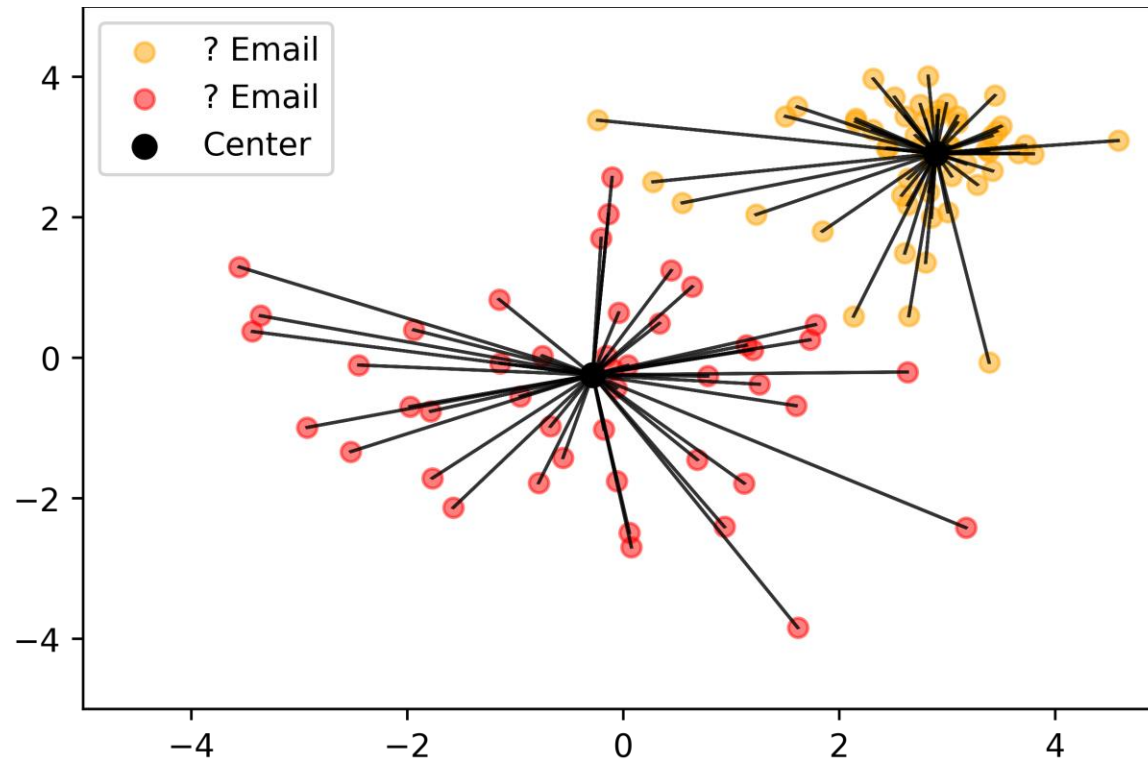
K-means clustering

- Uncovering “structure” in unlabeled data



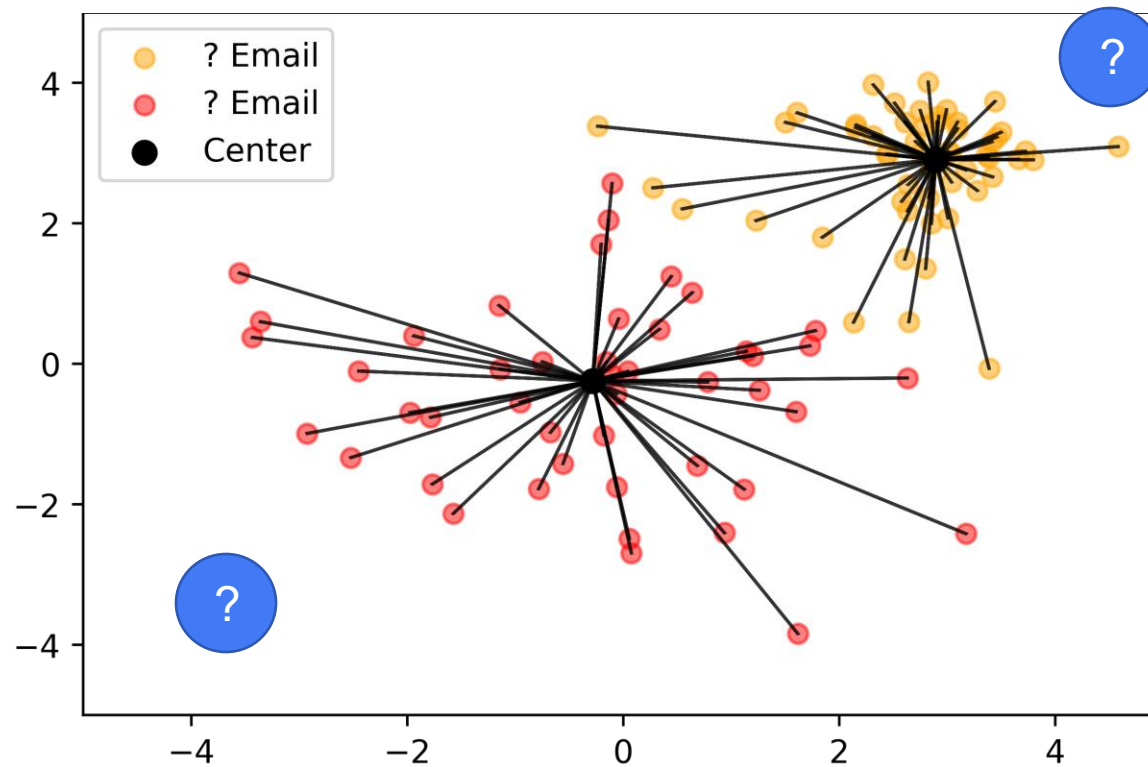
Clustering

K-means clustering



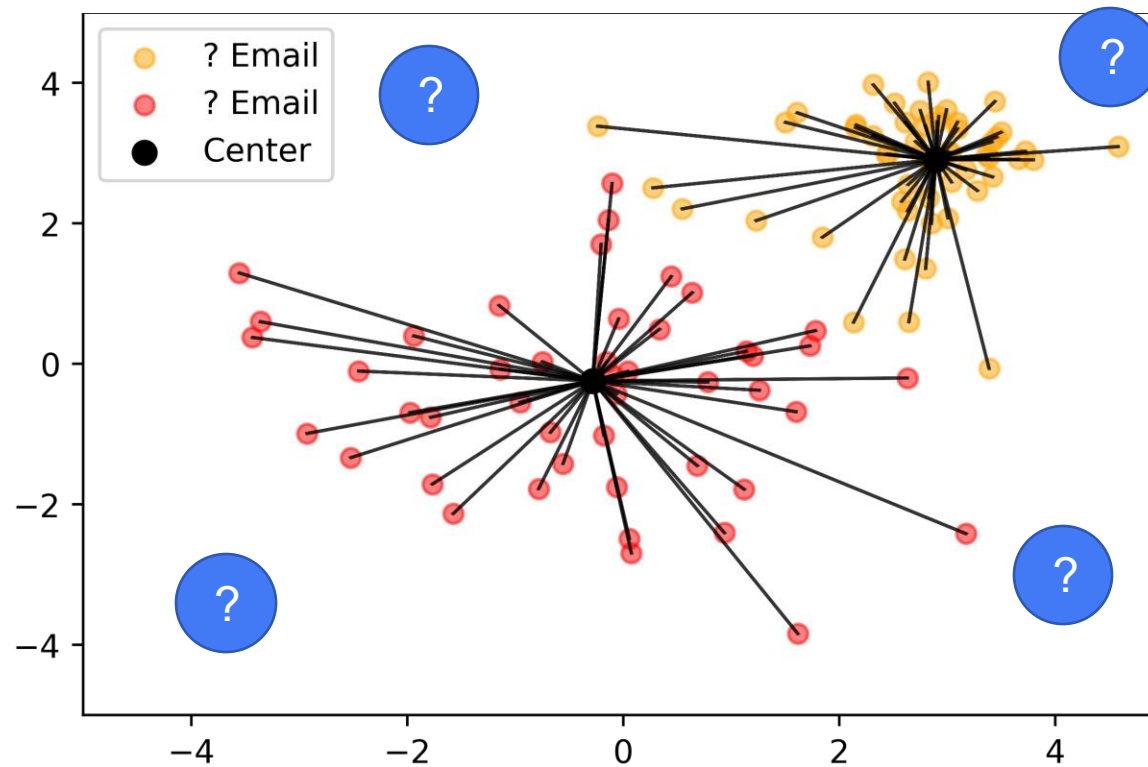
Clustering

K-means clustering



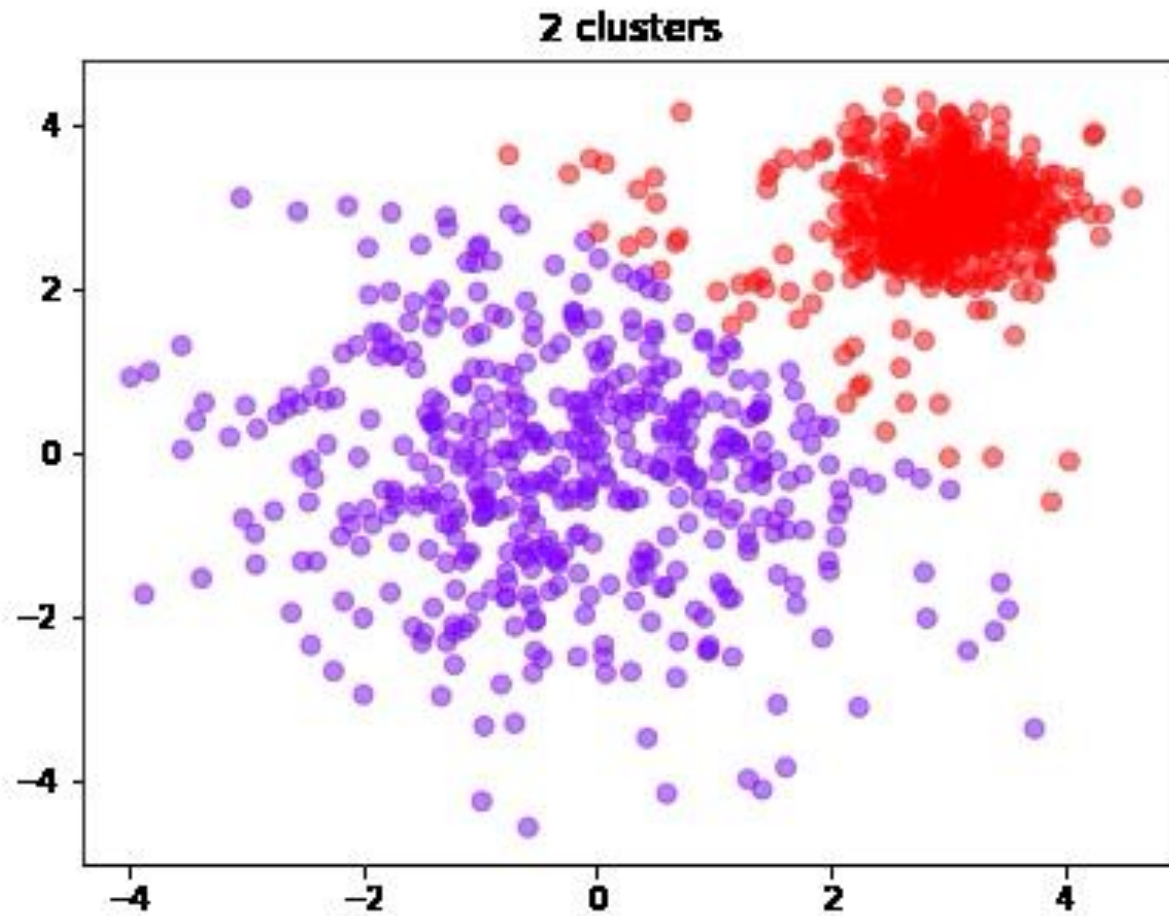
Clustering

K-means clustering



Clustering

Cluster Count?



Dimensionality Reduction

Unsupervised Learning

Dimensionality Reduction

Principal Component Analysis (PCA)

- Transforming high-dimensional data into a low-dimensional space

$$\begin{bmatrix} 4.5 \\ 2.3 \\ 2.4 \\ -99 \\ -3 \end{bmatrix} \Rightarrow \begin{bmatrix} 2.1 \\ 1.8 \end{bmatrix}$$

Dimensionality Reduction

Principal Component Analysis (PCA)

- Transforming high-dimensional data into a low-dimensional space

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$$

$$n > m$$

Dimensionality Reduction

Principal Component Analysis (PCA)

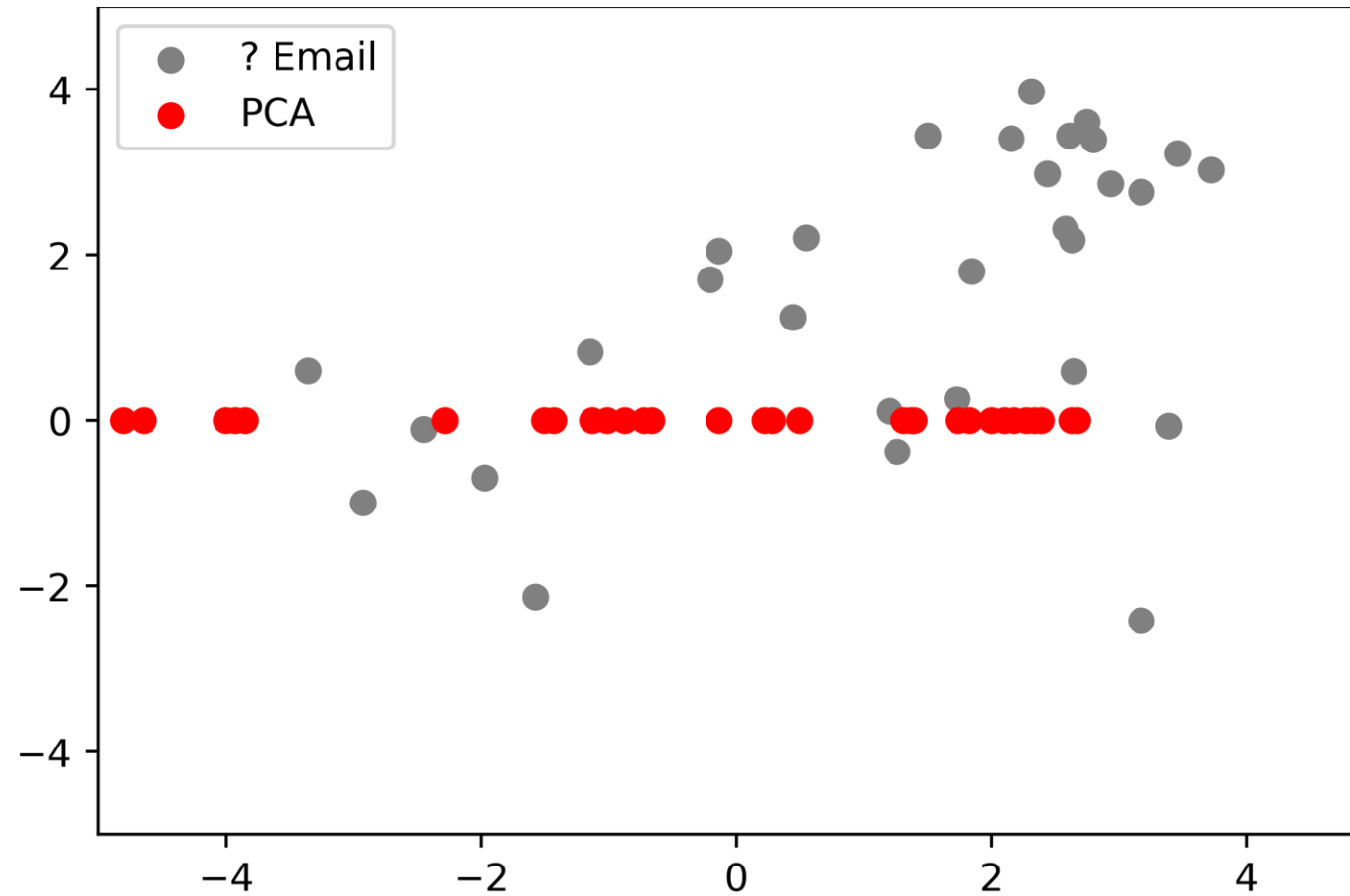
- Transforming high-dimensional data into a low-dimensional space

$$f_{PCA} \left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right) = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$$

$$n > m$$

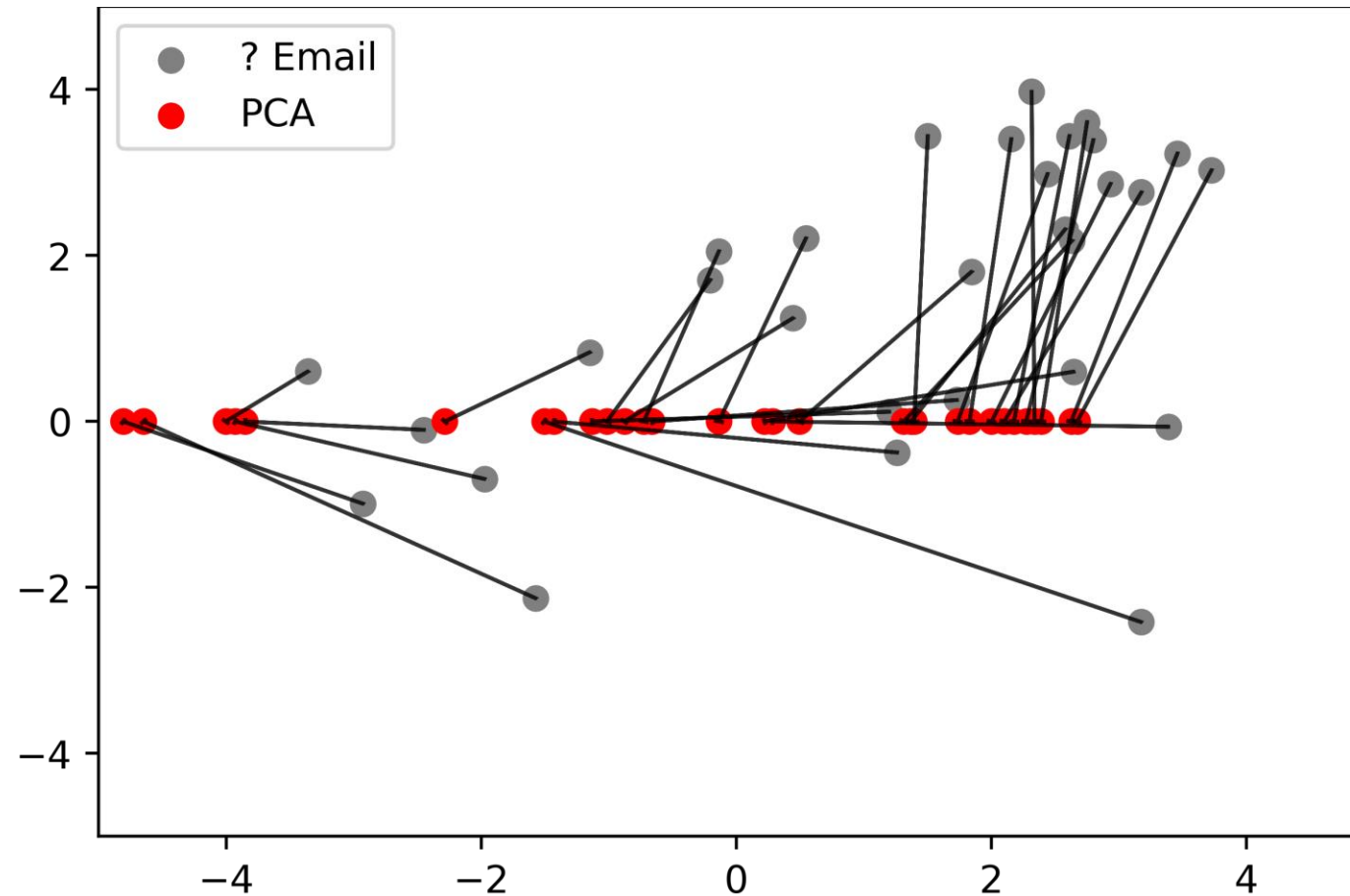
Dimensionality Reduction

Principal Component Analysis (PCA)



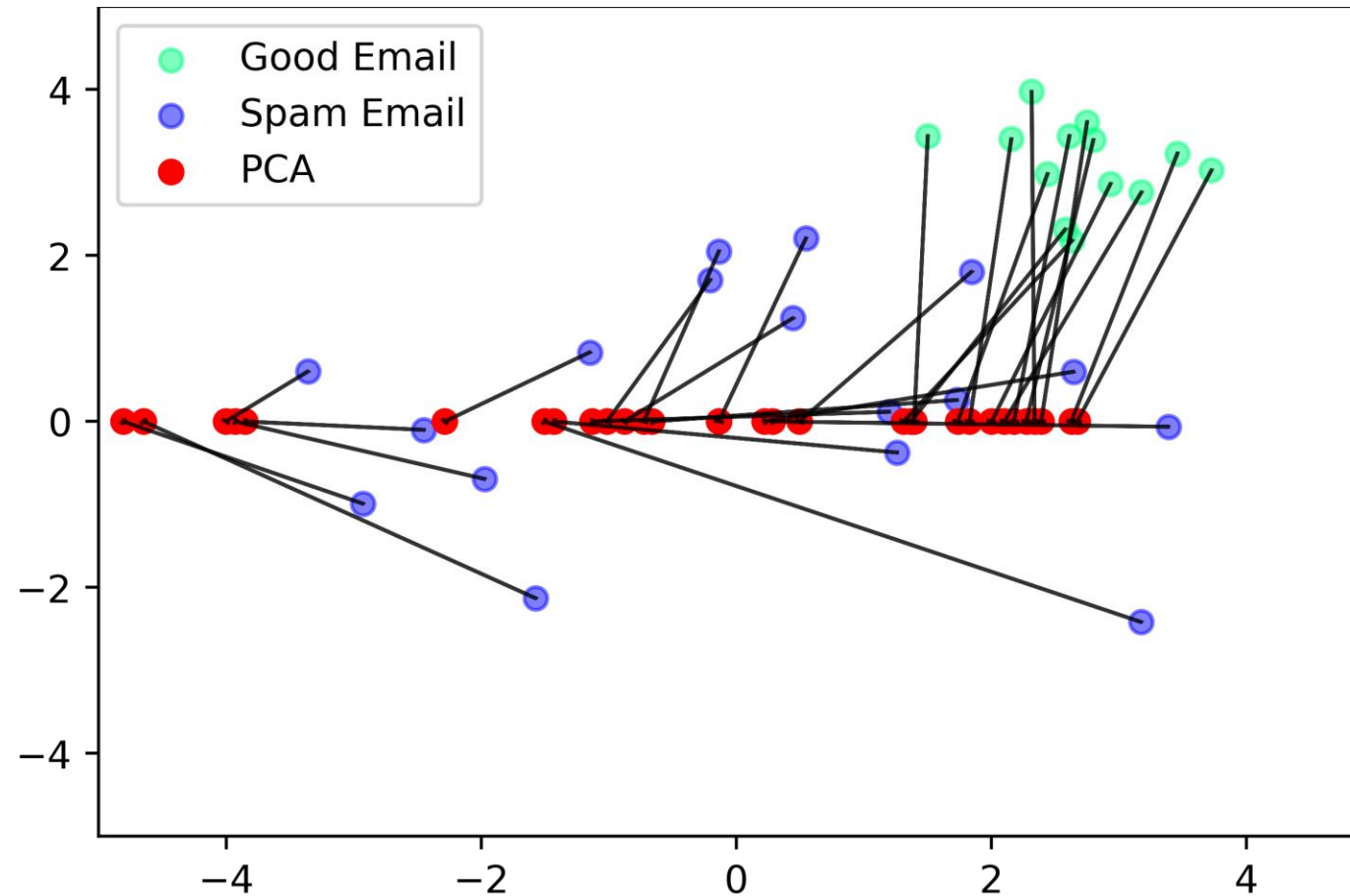
Dimensionality Reduction

Principal Component Analysis (PCA)



Dimensionality Reduction

Principal Component Analysis (PCA)



Conclusion

Unsupervised Learning

- Clustering
 - Finding meaning in unlabeled data
- Dimensionality Reduction
 - Transforming high-dimensional data into a low-dimensional space
 - Reducing the data to more essential features.

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